Southern Cross University

PHAR2001-2026-T5

Introductory Pharmacology

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Table of Contents

Classes

Unit Orientation - Part 1

- Welcome to the unit
- GenAI use guide
- Mandatory Modules at SCU
- Accessibility & technology support
- Your feedback
- Get ready for week 1

Unit Orientation Part 2: Pre-term activities

- A warm welcome from your Unit Assessor
- Let's practice learning from your interactive pharmacology textbook!
- Activity: Example 1 of the individual lessons: Scheduling of drugs
- Activity: Example 2 of the individual lessons: Medicinal formulations
- Activity: Example 3 of the individual lessons
- Activity: Start a folder for your notes!
- Your warm-up is finished: You are ready to go for it!
- Click on this video if you have ever said that you don't like online units

Assessments

- Assessment 1: The Quiz Folder
- Assessment 1 Brief
- Assessment 2: The workbook folder
- Assessment 2 Brief
- Assessment 3: The Case Study Folder
- Assessment 3 Brief
- The Interactive Pharmacology 'Textbook' involves all module content below

Module 1: Understanding Medications using the Gastrointestinal System

- Module overview
- Chapter 1.1: Medications that work by chemical or direct action

Introduction to Module 1: Chapter 1
Introduction to antacids
Activity: Safety issues of antacids
Specific antacids
Introduction to the laxatives
Understand constipation before suggesting laxatives
Osmotic laxatives
Stimulant laxatives
Activity: Mineral oil and detergent laxatives
Summary of Chapter 1
Chapter 1.2: Intro to Pharmacodynamics: Medications that work on cellular receptors
Introduction to Chapter 2
Understand safety issues around diarrhoea before treating it
Drug nomenclature (Naming)
Comparing Loperamide to other opioids
Cellular receptors
Affinity potency selectivity
Agonists antagonists and inverse agonists
Loperamide
Activity: Summary video
Chapter 1.3: Intro to Pharmacokinetics: Absorption, distribution, metabolism and excretion of drugs
Introduction
Absorption of drugs
Distribution of drugs
Metabolism of drugs
Excretion of drugs
Drug class affixes
Understanding the side effects of reducing stomach acid
Histamine 2 blockers
Activity: Watch this video summary of this chapter
Chapter 1.4: Medications that exert their effects on enzymes
Introduction

- Reviewing Pharmacokinetics by following omeprazole through the body
- Enzymatic inhibition by drugs, Part 1
- Understanding proton pump enzymatic inhibitors
- Understanding side effects of enzymatic inhibitors
- Omeprazole
- Summary of Chapter 4 on enzymatic inhibitors
- Chapter 1.5 CONDENSED (ONLY the ASSESSABLE material from: Medications that affect neurotransmitters
- Introduction to Chapter 5 on neurotransmitters
- Understand safety issues around vomiting before treating it
- Antiemetics for motion sickness
- Signing off of Module 1
- Chapter 1.5: Medications that affect neurotransmitters (FULL Chapter including non-compulsory lessons)
- Introduction to Chapter 5 on neurotransmitters
- Neurotransmitters
- Understand safety issues around vomiting before treating it
- Antiemetics for motion sickness
- Antiemetics for chemotherapy and GIT origins
- Activity: Watch this summary video

Module 2: The Respiratory and Autonomic Nervous Systems
- Module overview
- Chapter 2.1: Foundational information for understanding Autonomic Nervous System (ANS) Pharmacology
- Activity: Watch this video introduction to Chapter 1
- Overview of the Autonomic Nervous System (ANS)
- Divisions of the ANS
- Overviewing drugs of the ANS
- Selective receptors of the ANS
- Looking to the future
- Chapter 2.2: Sympathomimetics (drugs that mimic the sympathetic nervous system)
- Activity: Watch this video introduction to the chapter
- Anaphylaxis and adrenaline
- Other non-selective adrenergic agonists

- Selective alpha agonists
- Selective beta agonists
- Summary of Chapter 2
- Chapter 2.3: Sympatholytic Drugs (drugs that block the sympathetic nervous system)
- Activity: Watch this video introduction to the chapter
- Understanding sympatholytics
- Non-selective adrenergic blockers
- Alpha 1 blockers
- Beta blockers
- Summary of Chapter 3
- Chapter 2.4: Pharmacology of the Parasympathetic Nervous System (PNS)
- Activity: Watch this video introduction to Chapter 4
- Understanding acetylcholine
- Understanding the PNS
- Direct vs indirect cholinergic agents
- Indirect acting acetylcholine agonists
- Anticholinergic agents
- Summary of Chapter 4
- Chapter 2.5: Pharmacology of the Respiratory System
- Introduction to Respiratory System Pharmacology
- Preview of respiratory system
- Antitussives expectorants and mucolytic agents
- Decongestants and antihistamines
- Lower respiratory system disorders
- Treatment of Asthma (and other Ventilation Disorders)
- Congratulations on finishing the final Chapter in Module 2!

Module 3: The Endocrine System

- Module overview
- Chapter 3.1: Introduction to the Endocrine system, ADH, Oxytocin and GH
- Introduction to Endocrine System
- Review Part 1
- Review Part 2
- Oxytocin

ADH water saver (not compulsory for this unit)
Growth hormone (not compulsory for this unit)
Sign off chapter 1
Chapter 3.2: The Thyroid Axis
Understanding thyroid hormone
Hypothyroidism and its treatment
Hyperthyroidism and its treatment (Not compulsory)
Signing Off
Chapter 3.3: Diabetes and Metabolic Syndrome
Chapter 3.4: The Adrenal Axis
Introduction Stress Axis
Overview Stress Axis
Glucocorticoids
Side effects of corticosteroids
Signing off
Chapter 3.5: The Endocrine System and Bone Health
Introduction to medications for bone health
Understanding bone health
Understanding bone health 2
Bisphosphonates Teriparatide SERMS
Module 4: The Cardiovascular and Renal System
Cardiovascular System Chapter 1: Lipids
Introduction to Module 4 Chapter 1
Understanding Lipids
Understanding atherosclerosis
Understanding atherosclerosis, Part 2
The statins
Lowering cholesterol
Nicotinic acid
HDL levels
Summary
Cardiovascular System Chapter 2: Hypertension
Introduction to Module 4, Chapter 2

- Understanding blood pressure control, Part 1
- Understanding blood pressure control, Part 2
- Understanding blood pressure control, Part 3
- Hypertension
- The A, B, C, D's of the drug classes that are useful in hypertension
- Alpha Antagonists
- Medications that affect angiotensin II
- Alternative drugs for hypertension
- Calcium channel blockers
- Final words about antihypertensive agents
- Cardiovascular System Chapter 3: Thrombi
- Introduction to Module 4 Chapter 3
- Arterial clots
- Antiplatelet drugs
- Anticoagulants
- Fibrinolytics
- Summary
- The Renal System Chapter 4: Diuretics
- Introduction to diuretics
- Understanding diuretics
- Understanding diuretics 2
- Thiazide diuretics
- Loop diuretics (not compulsory)
- Potassium sparing diuretics (Not compulsory)
- Other diuretics (not compulsory)

Module 5: Analgesics and Psychotropic Drugs

- Module overview
- Analgesics
- Understanding Inflammation and pain introduction
- Mechanisms of inflammation and anti-inflammatory Agents
- Mechanisms of pain and analgesics
- Intro to fever and antipyretics
- Paracetamol (Acetaminophen)

- Pain relief in children and elderly
- Opioid overview
- Opioid mechanisms
- Specific opioids (Not compulsory)
- Chronic and neuropathic pain
- Psychotropic Medications
- Introduction to psychotropic medications
- Medications for Parkinson's disease
- Medications for treatment of Alzheimer's disease
- Anxiety
- Alternatives to benzodiazepines
- Depression
- SSRIs and TCAs
- SNRIs and atypical antidepressants
- Psychosis (Not compulsory for this unit)
- Specific antipsychotics (Not compulsory for this unit)
- Seizures (Not compulsory for this unit)

Module 6: Medications with Selective Toxicity

- Module overview
- Shortened Chapter: Only the 43 minutes of compulsory lessons on Medications with Selective Toxicity
- A/1: Antibiotics
- Antibiotics overview
- C/3: Antifungal agents
- D/4: Antiviral agents
- E/5: Anticancer chemotherapy
- Medications with Selective Toxicity (Full chapter, including non-compulsory lessons and activities)
- A/1: Antibiotics
- Antibiotics overview
- B/2: Specific antibiotics
- Specific Antibiotics 2
- C/3: Antifungal agents
- D/4: Viruses 1

Viruses 2
E/5: Anticancer chemotherapy
Polypharmacy 1: A reflective section
Acknowledgement of Traditional Custodians
Copyright notice

Classes

Timetable | Class Links | Class Recordings

Unit Orientation - Part 1

This module contains essential information about the unit. Read this information before Week 1 and refer to it regularly throughout the term.

Welcome to the unit

Unit overview

[Insert text here]

Unit Learning Outcomes (ULOs)

Welcome to PHAR2001, Introductory Pharmacology!

By the end of this unit, you will be able to:

Explain the main principles of pharmacokinetics and pharmacodynamics

Describe the mechanism of action, pharmacological effects, therapeutic uses, contraindications, safety issues and side effects of major drug classes and commonly used therapeutic agents

Discuss drug toxicity and interactions between drugs using specific examples of drugs and considering individual variability of drug response

Please engage in the pre-term activities (the final portion of the unit orientation, just prior to the modules) for introductory videos and an introduction to the way in which we will be learning in this unit.

-

GenAI use guide

Can I use GenAI in this unit?

No, unless it is specifically approved in the assessment brief. For this unit,

Assessment 1: quiz - NO AI AT ALL

Assessment 2 workbook: Formatting, spelling and grammar: Grammarly is permitted for language improvements only.

Assessment 3: To aid you in finding the best resources for answering the questions.

GenAI help resources

🔗 Generative AI for Students in the Southern Cross Model

🔗 How to acknowledge the use of GenAI in your work

🔗 SCU Library Guide: Gen AI

🔗 Learning Zone: Using generative AI tools

Can I use GenAI for assessments?

At Southern Cross University, you are permitted to use Generative AI tools to complete assessments when explicitly told to by your Unit Assessor.

For each assessment, your Unit Assessor will determine GenAI use level as one of the following:

- **Level 1: Full GenAI Use** - GenAI tools can be used in full throughout the assessment.
- **Level 2: Purpose-Specific GenAU Use** - GenAI tools can be used for specific assessment tasks or purposes as identified and scaffolded by the Unit Assessor.*
- **Level 3: No GenAI Use** - GenAI tools must not be used in any part of the assessment.

⚠️ For assessments classified Level 2, the Unit Assessor will specify the permitted and prohibited uses of GenAI within the relevant assessment brief.

Which GenAI tools can I use?

When approved for use in a specific assessment, Southern Cross University recommends that you use **Microsoft Copilot** with your SCU login. SCU has an enterprise agreement with Microsoft, which

Microsoft Copilot

🔗 Login to MS Copilot

means Copilot offers a higher level of data security and privacy than public, free-to-use GenAI tools.

You may also choose to use other widely available tools (e.g. ChatGPT, Google Gemini), but you do so at your own risk. SCU does not control or endorse third-party tools.

🔗 Copilot - GenAI - Library guide

How can I use GenAI safely?

Using GenAI can be helpful for your learning, but only if you use it safely and responsibly. Safe use means protecting your privacy, checking accuracy, and making sure GenAI supports (not replaces) your learning.

Your Responsibilities

- **Protect your data:** Never enter personal details, assessment questions, or sensitive SCU information into GenAI tools. Assume anything you type could be stored or reviewed.
- **Check accuracy:** GenAI can make mistakes, especially in technical areas like engineering materials. Always verify explanations, formulas, or examples against your lecture notes, labs, or trusted resources.
- **Respect assessment rules:** Follow the GenAI use level set for each assessment. Using GenAI in prohibited tasks counts as academic misconduct.
- **Own your learning:** Use GenAI to practise, test, or revise — but the real understanding must come from you.

Reduce Your Risk

- **Use Microsoft Copilot with your SCU login** where possible. It offers better security under SCU's enterprise agreement.
- **Check tool policies** before using external GenAI. Look for how your data is stored, whether it trains the model, and if you can delete it.
- **Minimise what you share:** Use made-up or anonymised data (e.g. fictional material strengths in practice problems).
- **Avoid full uploads:** Don't paste whole assignments or lab reports into GenAI. Work with short sections or sample problems instead.
- **Delete history:** Turn off chat history/model training where possible and clear conversations when finished.
- **Be transparent:** Keep a brief note of how you used GenAI, and acknowledge it if required in your assessments.

🔑 **Remember:** Safe use means GenAI should *help you learn* without taking away your opportunity to practise, demonstrate, and master core skills.

How can I show I've used GenAI responsibly?

The Unit Assessor may ask you to describe or demonstrate how you used GenAI and complied with the assessment guidelines. Be prepared to discuss which AI tools you used and how you used them.

⚠️ **IMPORTANT: All SCU students are required to:**

1. Record GenAI use

Keep a record of the Generative AI tools you used and how you used them. SCU's PAIR framework is a simple way to do this.

🔗 Quick Guide: The PAIR Framework

2. Acknowledge GenAI use

Correctly acknowledge and reference any Generative AI tools you use in your work. You'll find instructions on how to do this in these guides.

🔗 How to acknowledge the use of GenAI in your work

🔗 Quick Guide: Acknowledging and Referencing GenAI Use

3. Declare GenAI use

Before you submit any assessment task, you'll need to review the **GenAI Declaration**, which is available in the **Assessment Module**.

Mandatory Modules at SCU

Academic Integrity at Southern Cross University

Academic integrity is a core expectation for every student at SCU. Whenever you submit work, you are required to follow the **Student Academic and Non-Academic Misconduct Rules**. These Rules explain the standards for honest scholarship and the consequences if standards are breached. You are responsible for understanding these expectations and applying them in all assessments across your studies.

Not following the Rules can lead to serious penalties, including failing an assessment, failing the unit, or more significant University sanctions. If you ever feel unsure about what academic integrity means in your unit, speak with your Unit Assessor or visit the Academic Integrity website for more guidance.

Start the Academic Integrity Module

If you are starting your SCU studies, you must complete the **Academic Integrity Module** during your first study period. This module helps you understand academic integrity, the responsible use of sources, referencing, collaboration, and how to avoid common mistakes in academic integrity.

If you do not complete the module:

- Your grades may be withheld
- You may be withdrawn from future enrolled units
- You may be unable to enrol in further study

These penalties remain in effect until the module is successfully completed.

Completing the module usually takes **1 to 2 hours**, so it's best to complete it early in the session.

👉 **Click here to start the Academic Integrity Module**

Respect Matters?

At SCU, we're committed to building a community where everyone feels safe, respected and supported to learn, live and connect.

To support this shared understanding — and in line with the National Higher Education Code to Prevent and Respond to Gender-Based Violence — we've introduced a new module, Respect Matters. The module invites you to reflect on respectful relationships, consent, and how we can contribute to a safer, more inclusive SCU community.

Respect Matters replaces the previous *Respect at SCU* module and is part of how we support students to thrive at SCU.

To learn more about Southern Cross University's commitment to build safe and respectful communities visit Gender-Based Violence

👉 **Access the Respect Matters Module to get started**

Need Help?

For Respect Matters module enquiries contact Student Support Services on 1800 72 4357 (Press 3), or studentsafety@scu.edu.au

Accessibility & technology support

Accessibility support

SCU is committed to providing an inclusive and elearning environment for all. If you require accessibility services or support, please contact:

🔗 Inclusion Support

Technology support

If you encounter any issues using SCU's technology systems such as your student email or MySCU (Blackboard), please contact:

🔗 SCU Technology Services

Your feedback

How to provide feedback

Your feedback is incredibly important to us. Here are two ways you can share your feedback throughout the term:

Informal Feedback

For quick questions, suggestions, or immediate concerns, please feel free to reach out directly to your Unit Assessor or tutors via email or during workshops and tutorials. We're here to help.

Official Unit Surveys

Towards the end of the teaching period, provide confidential formal feedback via official university unit surveys. Your valuable input directly shapes the unit's future, so we'd love to hear from you.

You spoke, we listened

Your feedback helps us to make continuous improvements to our units. Here's how we have changed this unit for the better based on your insights:

You told us that...

The 40% end of term exam was too stressful.

...**So we changed** the exam to a 20% quiz in week 2 to make sure you have a good foundation with far less stress.

You told us that...

The case study was too long

...So we cut the length of the case study in half

You told us that...

The workbook was too big

...So we changed the workbook, reducing the number of multipart tutorial questions from 5 to 2

Get ready for week 1

PHAR2001 Introductory Pharmacology – Term 5, 2026

Week	Online Learning The online learning prepares you for the quiz and lays down foundational knowledge in pharmacology necessary for your future career. It is your interactive pharmacology textbook.	Online tutorial The tutorials are aimed at helping you do well in the assessments.	Assessment
0 (Orientation week)	**Orientation module**	This module prepares you to work most efficiently within the unit.	
1 26-30 October	**Module 1** 'Understanding medications' using medications of the gastrointestinal system: Pharmacodynamics, Pharmacokinetics and therapeutic foundations.	**Tutorial 1** Developing the research skills to succeed in the case studies (relevant to the case study)	
2 2-6 November	**Module 2** The respiratory and autonomic nervous system medications	**Tutorial 2** Quiz preparation	✓ **Assignment 1: Quiz (20%): Friday 06 November**
3 9-13 November **CENSUS DATE 9 Nov.**	**Module 3** Medications of the endocrine system	**Tutorial 3** Multi-part tutorial question A (relevant to tutorial workbook)	
4 16-20 November	**Module 4** Medications of the cardiovascular system	**Tutorial 4** Multi-part tutorial question B (relevant to tutorial workbook)	
5 23-27 November	**Module 5** Analgesics and medications of the central nervous system	**Tutorial 5** Multi-part tutorial question C (relevant to tutorial workbook)	✓ **Assignment 2: Workbook on tutorial questions is due Wednesday 25 November**
6 30 Nov -4 December	**Module 6** Anti-infective pharmacology	**Tutorial 6** Last chance to address case study questions	✓ **Assignment 3: Case study: due Wednesday 02 Dec.**
7 and beyond	*No further material*		*Grades will be published on 16 Dec.*

Unit Orientation Part 2: Pre-term activities



Welcome to the last phase of your unit orientation. Please take time to go through this very quick warm-up, so that you can save time and frustration.

A warm welcome from your Unit Assessor



Video: Welcome to PHAR2001

Hi All,

In this video, I give you a quick 2.5 minute overview of the unit and introduce myself; please join me by watching this first video.

Regards, Ken

https://youtube.com/embed/sZmdtFvNfCg

https://youtube.com/embed/JLbrwgDiDyE

calendar T5 2026.docx

Video: Your Term 5 study calendar

Click onto the YouTube play tab to watch the Mediasite quick video overview of the T5 2026 calendar and the unit. Download your own copy of the calendar by clicking onto the calendar document.

Video: Overview of the assessment structure in this unit

Please watch this video to get an overview of the assessments in this unit and to get a general idea of how to do well in the unit. (This is the same Overview of Assessments that is in the Assessment Tasks and Submission folder)

https://youtube.com/embed/zg41cEhK94k

Let's practice learning from your interactive pharmacology textbook!

Video: The first three lessons of your interactive textbook

This video introduces the first three lessons in your interactive pharmacology textbook.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

Please engage in the following lessons

The notes for the three videos that follow this introduction are just below.

Download the notes if they work for you and then simply click on the video to play.

Click below for the notes for the first three lessons and begin your folder of notes that will be invaluable in the final exam.

To maintain academic integrity in the exam given that the final exam is online, all questions in the exam must be worded in a way that students must look up 5 different answers in order to find the correct answer for each question. Students will not have the time to do this for more than a couple questions, so a much better approach will be to make notes on the notes provided and refer to your notes during the exam.

Activity: Example 1 of the individual lessons: Scheduling of drugs

Video: Scheduling of drugs

Please watch this quick video and engage in the drag and drop exercise below. Please note: in the video, I was not entirely clear in the fact that Alprazolam (Xanax) and flunitrazepam are the only two diazepines that are Schedule 8 at this time.

Just as people will **read** at different paces, you will find that you may want to **adjust the speed of the video** . If so, wait till it starts, hover on to the **1X** symbol, and other speeds will become available.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

scheduling of drugs 2

Remember that the information in the videos and activities within Module 1 is assessed in the 20% Week 2 Quiz. Make sure that you read the feedback that comes up with some of the answers.

Activity: Example 1 of the individual lessons: Scheduling of drugs

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_0_1_1_Scheduling_of_drugs

Activity: Example 2 of the individual lessons: Medicinal formulations

Video: Medicinal formulations

Please watch this two minute video, access the table and engage in the **two** drag and drop exercises below.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_0_1_2_Medicinal_preparations

Activity: Example 2 of the individual lessons: Medicinal formulations

Click Quick check table of medicine formulations and ab..docx to access your table

Quick check table of medicine formulations and ab..docx

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_0_1_2_Medicinal_Preparations

Q: Did you notice that there were TWO drag the word activities above? Click the blue circle to the right for the second one.

Activity: Example 3 of the individual lessons

Video: Pregnancy categories

Please engage in this quick 2 minute video and the multiple choice question below.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_0_1_3_Pregnancy_categories

Activity: Example 3 of the individual lessons

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_0_1_3_Pregnancy_categories

Activity: Start a folder for your notes!

Video: Constructing notes for the Week 2 quiz

In this video, I give a suggestion on how to construct your notes for the Week 2 Quiz.

https://youtube.com/embed/xUMeRnmpvUc

Example questions

Example of a question from this Orientation module that may be seen in the week 2 quiz on Module 1:

A medication that has been determined by the TGA to need to be dispensed by a pharmacist but a prescription is not necessary is classified as a:

1. Pregnancy category B medicine
2. Schedule A medicine
3. Schedule 2 medicine
4. Schedule C medicine
5. Schedule 3 medicine

Click below for the notes for the first three lessons and begin your folder of notes that will be invaluable in the week 2 quiz.

To maintain academic integrity in the quiz, all questions in the exam must be worded in a way that students must look up 5 different answers in order to find the correct answer for each question. Students will not have the time to do this for more than a couple questions, so a much better approach will be to make notes on the notes provided and refer to your notes during the quiz.

Your warm-up is finished: You are ready to go for it!

https://youtube.com/embed/2JcPi8Sw2Bw

Video: Ready to hit the ground running

Please remember that the prior three lessons are assessable in the week 2 quiz.

Please watch this final video of the orientation module, and then you are off and running!

Click on this video if you have ever said that you don't like online units

Video: Have you disliked any other online units?

I am pretty sure that you are going to change your opinion about online units after you engage in this one. Watch this video to find out why.

But if you still do not want to engage in the online version of Introductory Pharmacology,

JOIN US IN TERM 1 or 3 as an on campus student ANY YEAR!

https://youtube.com/embed/XK6X4cx5bgw

Assessments

Assessment Information| Assessment Briefs & Submissions | Assessment Help & Support

Assessment 1: The Quiz Folder

Assessment 1 Brief

Assessment 2: The workbook folder

Assessment 2 Brief

Assessment 3: The Case Study Folder

Assessment 3 Brief

The Interactive Pharmacology 'Textbook' involves all module content below

Module 1: Understanding Medications using the Gastrointestinal System



Module 1 lays down the foundation for your understanding of medications, your ability to use them safely and your ability to streamline your studies in Pharmacology. Using one of the easiest bodily systems, the gastrointestinal tract/system (GIT), we explore pharmacokinetics, pharmacodynamics and other foundational information. This is a very important module that is twice the size of the average module in this unit. The week 2 quiz on Friday of week 2 tests your knowledge of the module.

Module overview



Module 1: The basics of Pharmacology

Hi All,

Module 1 explores most of the important basics of Pharmacology to give you a solid foundation for your understanding of medications in the future. Normally, the foundational information in pharmacology is the most difficult (and sometimes boring) material.

But **in this module, we use the medications of the gastrointestinal system/tract (GIT) to explore these difficult topics, making it actually fun.** We start out with the easiest of medications - those that cannot get absorbed and work by chemical action or by direct action. (Normally, our medications work on cellular receptors or affect enzymes, but the first topic/chapter introduces you very slowly.) Next, Topic/Chapter 2, we explore cellular receptors, looking at what they are, what they do and how our drugs can affect them. This is where we get into the topic of pharmacodynamics. The third topic/chapter starts introducing

you to pharmacokinetics, the movement of drugs through the system, involving absorption, distribution, metabolism and excretion. We look at enzymes and how we can affect them with our medications in Topic/chapter 4. Finally, in Topic/chapter 5, we look at neurotransmitters and the most difficult concepts in the GIT, which have to do with our antiemetics.

If you want to love pharmacology the way I do, **please pay close attention to this first module.** I teamed up with a partner in this module to make sure that you can actually picture the processes that will form the basis of your understanding of pharmacology. Unfortunately, while all of the videos in this unit are of relatively high quality, this module will have a significantly higher level of annimation than subsequent modules, but by then you will have developed the ability of picturing how our medications work !

Enjoy! Regards,

Ken

Module purpose

By the end of this module you will be able to:

- Explain the main principles of pharmacokinetics and pharmacodynamics
- Describe the mechanism of action, pharmacological effects, therapuetic uses, contraindications, safety issues and side effects of the most important medications of the GIT



A note about the size of this module

This entire module contains a total of 4 hours of video lessons plus activities that reinforce your knowledge and prepare you for the quiz. The average of all 6 modules is 2 hours of video lessons per module, so this is significantly bigger than any other module. While the online material is much bigger this week than other weeks, we do not have any tutorial questions or questions in the Case Study this week, so your work this week is in the modules.

Chapter 1.1: Medications that work by chemical or direct action

Introduction to Module 1: Chapter 1

Introduction

Normally, medications work on cellular receptors or by affecting enzymes. They go through the process of pharmacokinetics, which involves absorption, distribution, metabolism and excretion. So, understanding them is pretty difficult. In this topic/chapter, we are going to start out with the easiest medications of the digestive system, the antacids and the laxatives. Since these do not get absorbed and they work by chemical or direct action rather on cellular receptors or enzymes, they are in themselves, relatively easy to understand. But remember that as you go through these topics, you are learning the foundation for your understanding of all medications, so please follow the step-by-step process of going through all of the material presented here.

In this module, the only NOTES that are available are notes: three to a page with spaces for your own notes. In subsequent modules, PowerPoints and notes will be available.

Click directly below to access the notes, which are arranged in order of the way the material is presented, from the introductory video lesson to the summary lesson.

The importance of engagement

The obvious reason to engage in the videos and activities in this module is that **they are the way that you will most efficiently prepare for the 20% quiz** , but the videos and activities will also help in several other ways.

Video: Why engage in the quizzes after each video?

Watch this video to see how the process works to help your memory of the facts that are most important for your future career.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

Why engage in the quizzes?

Introduction to antacids

What are antacids?

Right now, we are learning how to study pharmacology in the most productive way. If we develop the best approach to studying medicines using the easiest medications, we can apply that to other more difficult medications and disorders.

Antacids are **indicated** foar the occasional relief of heartburn and **contraindicated** as the heartburn (or gastroesophageal reflux) continues for an extended length of time (becomes chronic). In this lesson, we will understand why that is, because what we don't want to do is memorise long lists of what medications are indicated for and contraindicated in.

So please don't think that the topic is not valuable just because it is relatively easy!

Chapter overview

This first chapter is pretty easy, so you may want to speed things up. In this module, I put most of the videos on YouTube, so turning up or down the speed is just a bit different to the way in which you will turn up the speed in other modules.

To adjust the speed of this video, wait until the video starts, and then **press the settings tab** (the cog wheel) and choose PLAYBACK SPEED, which gives you options from 0.25 (1/4th of the original speed) to 2.0 (twice the original speed). Simply choose the one that is most appropriate for you. Just as some people read faster or slower than others, no vidoe speed will be perfect for everyone, so use this feature to get the most for the least amount of time. After finishing the video, make sure that you engage in the activity below to reinforce that knowledge and get ready for the 30% quiz. **I only use information found in the videos and activities when I design the quiz!**

Video: Overview of antacids

https://live.scu.edu.au/Mediasite/Play/aa562fe2185441f9a04c59245f6a85381d

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_1_1_Overview_of_antacids

Activity: Safety issues of antacids

Video: Some of the safety issues of antacids

This unit highly weighs safety issues when working with medications and despite the fact that antacids are freely available over the counter, there are important safety issues associated with them. Learn about those and much more in this video and make that information into active knowledge with the associated activities.

https://www.youtube.com/watch?v=pJjbiYjH3Cg

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_1_1q

Specific antacids

Types of antacids

There are four types of antacids that are frequently used in Australia. But how do you know which is best for the individual? For instance, if client A is normally a bit constipated, they would do best by taking the magnesium antacids, and if client B generally has softer stools,

they may do better with the aluminium antacids. This lesson gives you a clear picture of the strengths and weaknesses of all four antacids so that you can decide which, if any, is best for the individual client.

Video: Specific antacids

Please watch this 5 minute video (or 2.5 minutes turned up to 2x speed) and then engage in the activities below.

https://www.youtube.com/watch?v=zrFF_3uBtcA

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_1_2

Introduction to the laxatives

Introduction to the laxatives

We continue with our exploration into the easiest of our medications with a look at the laxatives.

Laxatives are the number 1 most abused medications because they are OTC (over the counter) and clients often self-prescribe when they should just be consuming more water and eating more fibre. Interestingly, there are a number of safety issues about the use of laxatives, so in this portion of the topic/chapter, we will learn a lot about how to safely suggest a medication with relatively easy examples. Remember, we are not just learning about laxatives, as we will be applying the information that we learn to other medications, conditions and scenarios.

Video: Introduction to laxatives

This lesson begins the second half of the chapter, and as such I present the aims and general approach. Please watch the short 2.5-minute video and follow it up by quizzing yourself below. Remember that I made notes, 3 to a page to help you (all the way at the top of module), which you may find useful to jot down important points as you go through the videos. Writing things down has a really strong ability to reinforce knowledge.

https://www.youtube.com/watch?v=EMaG0ejl30E

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_1_3

Understand constipation before suggesting laxatives

Understanding constipation

Remember that we need to:

- know the general physiology of the area and
- aim to understand a bit about the disorder (the PATHO-physiology) that the medication we are studying is used for.

Before we can understand why our medications are being used, how they will work and in what circumstances they will be contraindicated. The alternative is to memorise long lists of drugs that we don't understand at all. While we are talking about constipation in this instance, which is relatively easy, we will apply these techniques to be able to understand much more difficult disorders and drugs.

This section allows you to understand constipation so that you can safely recommend the best therapy.

Video: Understand issues around constipation

Please engage in this video and activities.

https://www.youtube.com/watch?v=dtoCoythljM

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

understand issues around constipation

Osmotic laxatives

Osmotic laxatives

Can you really picture the process of osmosis? If so, that is great, you will understand the osmotic laxatives quite easily. If not, this video will help you picture the process as you learn about the osmotic laxatives with this lesson. One of the most important points about osmotic laxatives is the fact that they develop a **linear dose response.** You will find out what this is and why it is very uncommon with our medications and very dangerous to assume that our other medications have that type of response.

In the quiz at the end of the Term, I will definitely include a question about linear dose response and other material presented in this 6-minute video. Please watch it to learn more about osmosis, safety issues of osmotic laxatives, linear dose response curve, normal dose response of medications and even the safety issues of paracetamol. I cover so much important information in this first module, so please give each of lessons your best shot.

Video: Osmotic laxatives exhibit a linear dose response

https://www.youtube.com/watch?v=kwm8-GNfq8U

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_1_5_Osmotic_laxatives

Stimulant laxatives

Stimulant laxatives

If you have ever had a GIT 'bug' and ended up going to the toilet about 20 times, you may have wondered why that was happening. Well, the toxins from the bacteria and the bacteria themselves are basically working on the enteric nervous system of the GIT to signal an increased motility and increased secretions. We want to get those toxins out ASAP!

That is a simplistic picture of what the stimulant laxatives do. They irritate the GIT to really get the intestines moving.

But think about that. It is really logical when you think about them, that chronic (long term) use of the stimulant laxatives would result in damage to the GIT. Find out more about the stimulant laxatives and the very short list of situations in which they are indicated by participating in this lesson.

But wait, there is more! (there is always more in this module, so try not to skip anything). We talk about **enteric coating,** (why some drugs use an enteric coating, how to take drugs that are enteric coated, and what not to do with them) as well.

In the above description of this lesson, I note a number of important points that you will learn by watching this 7 minute video and engaging in the activities.

Video: Stimulant Laxatives: Sodium Picosulfate, Senna, Castor oil and Bisacodyl

https://www.youtube.com/watch?v=Z5R9v28VV0c

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_1_6_Stimulant_laxatives

Activity: Mineral oil and detergent laxatives

Mineral oil and detergent laxatives

What happens to something that gets into the circulatory system if we don't have a method to excrete it? Does it just keep circulating around and around or do we try to seclude it from the rest of the body in some way?

Go ahead, watch the video below to find out.

Video: Mineral oil and detergent laxatives: Serious health concerns when combined

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_1_7_Mineral_oil_and_detergent_laxatives

Summary of Chapter 1

Summary

Congratulations, as you have just finished the first chapter of Module 1.

This module is much bigger than subsequent modules in Introductory Pharmacology, but it has to be because we are learning all of the basics that will help you understand the material presented in subsequent modules. So on to the next chapter where we learn about the most common way our medications work, namely on cellular receptors.

Video: Understanding medications, chapter 1 summary

Watch this quick 2 minute summary of this chapter.

https://www.youtube.com/watch?v=2zlC9uQwHMU

Chapter 1.2: Intro to Pharmacodynamics: Medications that work on cellular receptors

Introduction to Chapter 2

Introduction to Pharmacodynamics

Now that we understand some of the foundational information about medications, we can move on to the more complicated topics of Pharmacology, namely Pharmacodynamics. Pharmacodynamics basically means 'what the drug does to the body' - how does it work?

Most of our medications work on cellular receptors, so it is important that in this chapter, we get a good, clear picture of what the cellular receptors are and how our medications can use those to exert their pharmacological effects.

The NOTES for this portion of the Module are directly below.

Video: Introduction to chapter 2

Watch this introduction to Chapter 2 of Module 1, and reinforce your knowledge with the activity.

https://www.youtube.com/watch?v=2-8Jm2LIQ0M

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_2_0_Introduction_to_Chapter_2

Understand safety issues around diarrhoea before treating it

Safety issues to consider

As we learned in the last topic/chapter, in order to SAFELY work with medications, we must first understand a bit about the disorder that is being treated. *Are there any safety issues ABOUT THE DISORDER that we need to be aware of before we jump in and recommend anything?*

So, since the underlying medication that we will be investigating in this chapter is loperamide, which is indicated for diarrhoea, our first job is to make sure that we understand the safety issues around diarrhoea. Right now can you think of any? What age groups and body shapes do you think would be most susceptible to those issues?

Video: Understanding safety issues around diarrhoea

You can get a pretty good idea of how important a lesson is by the number of activities. Watch this video and test your knowledge of the topic with the four activities below. I place a very high emphasis on safety, because that **is** the most important thing about working with meds.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_2_1_Understand_Safety_Issues_Around_Diarrhoea_Before_Treating_It

Drug nomenclature (Naming)

Naming conventions

There are many different names for the same medication, so do we need to learn them all? Are there any tricks that we can learn to help us remember the most important drugs?

While we will no doubt need to know the common brand names of many of our medications, it is the generic name that is most important in the first instance, because that is the one that is always on the label. There are affixes (usually suffixes) to start looking out for and this video attempts to get you in the habit of finding that suffix, and identifying the drug.

Video: Drug nomenclature (naming)

After you watch this video and do the associated activities, you will never look at a drug name the same way that you do now!

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_2_2_Drug_Nomenclature_(Naming)

Comparing Loperamide to other opioids

Understanding Loperamide

We have all heard about the opioids. The drug class that includes some of the strongest painkillers that we have. But Loperamide is an opioid but in normal doses, it has not pain-

relieving properties. So why is that?

We compare Loperamide to other opioids in this lesson so that you can get an overall picture of the drug class called the opioids.

We also investigate an important safety issue about loperamide: remember what we said, that 1/3rd of 1% of the medication gets into the system? Some people try to get an opioid high from consuming several times the recommended dose. Find out what happens in this lesson.

Video: Comparing Loperamide to other opioids

Despite the fact that an average of 0.4% of loperamide gets absorbed, there are important safety issues that you will learn about as you engage in this video. Remember that you can speed up or slow down the videos by clicking on the settings tab, and choosing a setting different to 1.0.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_2_3_Comparing_Loperamide_to_Other_Opioids

Cellular receptors

Importance of cellular receptors

These next three lessons are absolutely essential for you to understand if you intend to understand our medications. I deleted a lot of the detail about cellular receptors so that you can concentrate on really understanding the most important points. Try to imagine the receptors in this lesson. No need to memorise the types for this unit - just try to get that overall picture.

Video: Cellular receptors

Please don't miss this one. Watch how our medications bind to cellular receptors and what happens after binding, then reinforce your knowledge with the three activities below.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_2_4_Cellular_Receptors

Affinity potency selectivity

Understanding affinity, potency and selectivity

Why is it that medication A, which works in the same way as Medication B can be so much stronger (more potent)? While there can be many reasons, one of the most common has to do with the strength of the molecular bond of the medication to its given receptor.

Most often, a medication that affects cellular receptors binds to its receptor and then lets go shortly after. Then it binds to another receptor of the same type, and lets go... But what if the drug bound to the receptor and it really takes a lot more time to get it off that receptor (it had a much greater affinity to the receptor)? As you can imagine, many more messages will be sent to the cell. So the drug's affinity can affect its potency.

This is another really important lesson as we learn three very important concepts, affinity, potency and selectivity.

Video: Affinity potency selectivity

The terms you learn by watching this video will be used throughout this unit. Make sure that you engage in the activities directly after watching the video.

(Please also note the two questions within the video associated with this lesson)

https://www.youtube.com/watch?v=QTRj5_KZ1NQ

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_2_5_Affinity_Potency_Selectivity

Agonists antagonists and inverse agonists

Video: Agonists, antagonists, inverse

agonists

There are many different ways that a drug can affect a particular receptor. This lesson gives you an overview that you can really picture.

I cannot stress how important these concepts are. Please engage in the video and activities below.

Activity: Test your knolwedge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_2_6_Agonists_Antagonists_and_Inverse_Agonists

Loperamide

Video: Loperamide

After engaging in that foundational information that will help you understand any medication that is an agonist, you can now easily understand loperamide with this lesson.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_2_7_Loperamide

Activity: Summary video

Video: Summary of chapter 2

This video summarises Module 1, Chapter 2.

After watching this 3-minute video, give yourself a big pat on the back, because you are starting to develop a really good picture of medications that will help you describe medications and their effects to your clients and colleagues in the future.

Also remember that these videos and activities prepare you for the workbook questions and also are the best preparation for the end of Term quiz, so well done on completing this second chapter in Module 1!

Chapter 1.3: Intro to Pharmacokinetics: Absorption, distribution, metabolism and excretion of drugs

Introduction

Video: Introduction to Pharmacokinetics

Welcome to Module 1 chapter 3 where you will begin understanding how drugs move around the body. The process of pharmacokinetics, which involves absorption, distribution, metabolism and excretion. I quickly introduce this topic in this 3 minute video.

https://www.youtube.com/watch?v=9gUUP5L8_kc

Absorption of drugs

Video: Absorption of drugs, Part 1

In this lesson, we begin to understand the advantages and disadvantages of different modes of administration as we truly picture them.

Important stuff. Please engage in the 6 minute video, which has a couple reflective quizzes at the end, and then engage in the two activities below.

Activity: Test your knowledge, Part 1

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_3_1_Absorption_of_Drugs_Part_1

Video: Absorption of drugs, Part 2

The second half of this video includes a couple questions that will help you apply the information from the first half of the video. Please engage in those questions.

Activity: Test your knowledge, Part 2

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

Clone of PHAR2001_1_3_2_Absorption_of_Drugs_Part_2

Distribution of drugs

How drugs are distributed within the body

The distribution of lipophilic drugs involves protein binding which is a very difficult concept to understand but this video will really get you to picture the process.

Please engage in the video and in the two exercises that are presented in the second half of the video, and then in the Activity: Quizzes for this lesson, in that order.

Thanks, Ken

Video: Quizzes for this lesson

I would have loved to have learned about pharmacokinetics in this way. Please engage in the video and associated activities.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_3_3_Distribution_of_Drugs

Metabolism of drugs

The importance of the liver

The liver is the site of most metabolism of drugs. As soon as most drugs are absorbed and begin the distribution process, our body wants to excrete the drug as quickly as possible. And since the kidneys are the main site of excretion of most drugs, the main goal of metabolism of drugs is to make the drug more water soluble and easy to excrete. We want that drug out of there ASAP

Interestingly, as drugs go through the liver, many different scenarios are possible, including that an inactive pro-drug can become an active drug or an active drug can become an inactive drug. This lesson explains the basics of the third principle of Pharmacokinetics, the metabolism of drugs, in a way that you can apply the information to any medication.

Video: Metabolism of drugs

Please watch this video and engage in the three activities in the following sections.

https://www.youtube.com/watch?v=naal1V-0iaw

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_3_4_Metabolism_of_Drugs

Excretion of drugs

How drugs are excreted

The main thing that we need to learn about with respect to the excretion of drugs has to do with the half-life. Half-life is defined as the time required for any specified property (e.g. the concentration of a substance in the body) to decrease by half. Interestingly almost all of our prescription medications are eliminated differently to that of ethanol, the alcohol in alcoholic beverages.

Video: Excretion of drugs

I have condensed the excretion of drugs so what is left (5 minutes) is really important. Please engage in this lesson.

https://www.youtube.com/watch?v=5yA4u8q96I4

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_3_5_Excretion_of_Drugs

Drug class affixes

Drug name classifications

Have you ever looked at the label on a container of pills and your eyes twisted the name in all sorts of ways to get the generic name make sense? It never does. ...or does it? If you learn a simple trick contained in this lesson, you will be able to identify general properties of drugs you have never seen before.

Video: Drug class affixes

You may be wondering how you can learn quite a bit about 22 of the 200 most prescribed medications in 6 minutes and 17 seconds. Well, watch the video and find out! Don't forget to engage in the drag and drop excecise because that will help reinforce your knowledge.

BTW: Ranitodine was mentioned here but has been withdrawn from the Australian market. Other ...tidine mentioned are all available.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_3_6_Drug_Class_Affixes

Understanding the side effects of reducing stomach acid

The importance of stomach acid

Histamine 2 blockers (the drug class with the ...tidine suffix) do not block all of the acid from the stomach like proton pump inhibitors (the ...prazole drugs) do, but nonetheless, they have similar long-term side effects associated with the greatly reduced stomach acid.

If you think about it, the acid is there for a reason. And some clients will be blocking that acid production for years. So let's quickly review the functions of the acid in the stomach and see how that ties into the side effects that are likely from long-term decreases in stomach acid.

Video: Understanding side effects of reduced stomach acid

Millions of Australians use these medications that are available without a prescription and without knowing the potential side effects associated with their long term use. Understand those effects in just 5 minutes with this video, and you will be able to help clients or friends who decrease their GIT acidity

without getting the problem properly assessed.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_3_7_Understanding_the_Side_Effects_of_Reducing_Stomach_Acid

Histamine 2 blockers

Safety issue

The most popular Histamine 2 receptor blockers (due to better efficacy and potency) was Ranitidine (Zantac) until 2020 when it was discovered that some components of the drug broke down into carcinogenic components. This happened to a greater extent over time and if the drug was stored improperly, like in a hot car. The company that manufactured Zantac has changed the drug from Ranitidine to Famotidine and changed the brand name of their version of Famotidine to Zantac 360. Clients who mention having Zantac should be questioned to make sure they are not taking leftover Ranitidine.

Video: Histamine 2 receptor blockers

All other Histamine 2 blockers like Famotidine and Cimetidine and other ...tidine drugs have been tested for carcinogens and none were found. This video excludes mention of Ranitidine and attempts to give you a good picture of the other members of the Drug Class

https://www.youtube.com/watch?v=B_WPSws-eYg

called the Histamine 2 receptor blocker drugs.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_3_8_Histamine_2_Blockers

Activity: Watch this video summary of this chapter

Chapter summary

Well done on completing this chapter. Hopefully you have enjoyed developing a picture of the process of pharmacokinetics. If you do not feel like you caught it all the first time around, don't worry because in the next chapter, we follow the main drug we discuss, as it goes through each process of pharmacokinetics: absorption, distribution, metabolism and excretion.

Thanks for giving this difficult module your best shot, well done. I spent a lot of time trying to get you to picture these complex processes, and it is all worth it when I see you benefiting!

Regards, Ken

Video: Chapter 3 summary

https://www.youtube.com/watch?v=aNs11g8RgnM

Chapter 1.4: Medications that exert their effects on enzymes

Introduction

Introduction to Chapter 4

Welcome to Module 1 Chapter 4, where you will begin understanding enzymes and how our medications affect enzymes. In this chapter, we use the proton pump inhibitor to understand all of the foundational information about enzymes and the effect of drugs on them.

Here are the NOTES, three to a page with room for your own notes.

Module 1 Chapter 4 NOTES 2025.pdf

Video: Introduction to Chapter 4 - Enzymatic inhibitors

https://www.youtube.com/watch?v=IZZq4VdfpRs

Activity: Test your knowledge

Please engage in this activity to make sure that you understand the material in the introduction video.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_4_0_Introduction to Chapter 4 on Enzymes

Reviewing Pharmacokinetics by following omeprazole through the body

A review of Pharmacokinetics

Before we get on to the discussion of enzymes and enzymatic inhibitors that are the main topic of this chapter, **let's take a review of the last chapter's topic, but using the drug we will be discussing in this chapter. The material in this video is a review and as such is not mandatory, but make sure that you engage in the activities, as those are assessable.**

So in this lesson, we use Omeprazole, (a proton pump inhibitor that decreases stomach acid) to review all of the processes involved in Pharmacokinetics: absorption, distribution, metabolism and excretion. You don't need to memorise anything, just sit back and watch the drug flow through the system as you review the processes involved in one of the most difficult topics in Pharmacology.

Video: Pharmacokinetics - Visualising Omerprazole

This is an interesting review of the four processes of pharmacokinetics, using omeprazole as the drug. Sit back and watch it get absorbed, distributed, metabolised and excreted. **This video follows a drug (omeprazole) through the body as a review of pharmacokinetics, and as such it is not compulsory but make sure that you engage in the activities directly below, which cover a couple points that were not discussed elsewhere.**

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_4_1_Reviewing Pharmacokinetics by Following Omeprazole Through the Body

Video: Applying your knowledge quiz on Omerprazole

The video takes new material and as such is compulsory

In the review of Pharmacokinetics above, I left you thinking about a couple questions:

- What happens to the other 20% of the Omeprazole that does not get into systemic circulation?
- If the half-life of Omeprazole is 1.5 hours, why does it work in the system for up to 72 hours?

After thinking about those questions, watch this video to picture what actually happens.

Enzymatic inhibition by drugs, Part 1

Learning about enzymatic inhibition

Why is it that irreversible enzymatic inhibitors like aspirin and NSAIDs affect platelets for the life of the platelet (9 days) but only decrease inflammation for four hours? How can our drugs affect enzymes? So much to learn about enzymes and how our medications affect them but learning about it is interesting.

Video: Enzymatic inhibition by drugs, Part 1

Get a really good picture of the second most important way in which our medications work by clicking onto the video.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_4_2_Enzymatic_Inhibition_by_Drugs_Part_1

Understanding proton pump enzymatic inhibitors

Video: Understanding proton pump enzymatic inhibitors

In order to really understand how Omeprazole works, you need to understand the complex enzyme called the hydrogen-potassium ATPase pump. It is a pump and an enzyme and that makes is so very interesting!

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_4_3_Understanding_Proton_Pump_Enzymatic_Inhibitors

Understanding side effects of enzymatic inhibitors

The side effects of enzymatic inhibitors

How is it that something like grapefruit juice (and many of our drugs) can make some drugs more potent and other drugs less potent? How can drugs 'induce' some enzymes while inhibiting others? In this lesson, we continue using the proton pump inhibitor Omeprazole to learn about enzymes and how drugs can affect enzymes.

Video: Understanding enzymatic inhibitors

Enzymatic inhibition is the second most important way in which our medications work, so continue your understanding about enzymatic inhibition by engaging in this video lesson.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_4_4_Understanding side effects of enzymatic inhibitors

Omeprazole

Video: Omeprazole

Omeprazole is a proton pump inhibitor that is nearly 100% effective in reducing stomach acid and as such is the first line therapy for peptic ulcer disease and GORD/GERD. Pregnancy categories are also briefly reiterated as Omeprazole is a pregnancy category B3 drug.

Please watch this video and engage in the activities below.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_4_5_Omeprazole

Summary of Chapter 4 on enzymatic inhibitors

Video: Summary of chapter 4

I summarise the chapter in this final video.

You are really close to finishing the module, just a few videos in Chapter

https://www.youtube.com/watch?v=9mBIKv06txE

5.

See you there!

Ken

Chapter 1.5 CONDENSED (ONLY the ASSESSABLE material from: Medications that affect neurotransmitters

Introduction to Chapter 5 on neurotransmitters

Introduction to neurotransmitters

This is the condensed version (assessable portions only) of the last chapter of Module 1, and deals with drugs that work to control emesis or vomiting.

The aims of this chapter are to:

- Identify the safety issues associated with vomiting
- Explain the four pathways involved in emesis
- Place our medications into the context of the above

Please feel free to use the NOTES for the condensed Chapter 5 directly below. If you would like the full notes, please go to the full chapter 5.

Understand safety issues around vomiting before treating it

Understanding which antiemetic to administer

You probably have this down pat by now, and that is good, because it is important. If you do not understand the basics of the disorder that is being treated, then you may miss heaps of **red flags** and that could wind up in harm to the client.

So right now, before we get into the actual antiemetics, let's get a good picture of the types of safety issues that are involved with vomiting itself.

Video: Determine the underlying cause

The most important thing we need to do when working with clients is to BE SAFE. Ensure that you know the safety issues around vomiting by engaging in this 4.5 minute video.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_5_2_Understand safety issues around vomiting before treating it

Antiemetics for motion sickness

Treating motion sickness

Since there are four completely different pathways to the vomiting centre, using four distinctly different sets of neurotransmitters, emesis (vomiting) needs to be treated with the correct antiemetic.

This lesson helps you understand which antiemetics must be used to address motion sickness and which ones should be used for the type of vomiting that originates from the higher brain centre (like the vomiting that occurs when you think "looking at that just made me sick"').

Video: Antiemetics for motion sickness

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

Condensed version of antiemetics

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_5_3_Antiemetics for Motion Sickness

Signing off of Module 1

Conclusion of module 1

Congratulations on completing the most difficult topics in Pharmacology!

Please reward yourself for giving this module your best shot. You will now understand the material in subsequent chapters, and most of the discussions about medications that you will engage in with colleagues, and you will be able to explain medications to clients. You have also begun your preparation for the 30% end of Term quiz.

See you in Module 2!

Warm regards, Ken

Chapter 1.5: Medications that affect neurotransmitters (FULL Chapter including non-compulsory lessons)

Introduction to Chapter 5 on neurotransmitters

Introduction to Neurotransmitters

You do not need to engage in this full chapter for PHAR2001. You may use the condensed chapter adjacent to the point at which you enter this file.

This is the last chapter of Module 1, and deals with drugs that work on neurons in general and specifically, work to control emesis or vomiting.

The aims of this chapter are to:

- Generally explain neurotransmission and the ways that our medications can work on neurons
- Identify the safety issues associated with vomiting
- Explain the four pathways involved in emesis
- Place our medications into the context of the above

Please feel free to use the NOTES for Chapter 5 directly below:

Module 1 Chapter 5 NOTES 2025.pdf

Video: Watch this video introduction to the chapter

Bismuth subsalicylate is not sold in Australia at all, and as such, there will be no questions in the quiz regarding that medication. Please just aim to be aware of its indications and effects, and the fact that it is not available here, because it is available in just about every other country, and is heavily relied

on by many people who have frequent stomach irritation.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_5_0_Introduction to Chapter 5 on neurotransmitters

Neurotransmitters

What are neurotransmitters?

Neurons communicate with the adjacent neuron using a neurotransmitter that excites or inhibits the next neuron. But we need to get that neurotransmitter back out of the area very quickly by a number of methods, so that there is not constant excitation or inhibition at the next neuron. Our medications that affect neurotransmission can target the receptors on the neurons, they can mimic the neurotransmitter, or they can target the ways in which we get the neurotransmitters out of the synaptic cleft. The antiemetics all target the receptors on the neurons. In other words, they are receptor antagonists for the neurotransmitters involved in emesis.

Video: Understanding neurotransmission

We discuss neurotransmission in Module 5, but it would be good to know some of that information as we discuss antiemetic drugs right now.

Activity: Test your knowledge

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PHAR2001_1_5_1_Introduction to Chapter 5 on neurotransmitters

Understand safety issues around vomiting before treating it

Safety issues with vomiting

You probably have this down pat by now, and that is good, because it is important. If you do not understand the basics of the disorder that is being treated, then you may miss heaps of **red flags** and that could wind up in harm to the client.

So right now, before we get into the actual antiemetics, let's get a good picture of the types of safety issues that are involved with vomiting itself.

Video: Determine the underlying cause

The most important thing we need to do when working with clients is to BE SAFE. Ensure that you know the safety issues around vomiting by engaging in this 4.5 minute video.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_5_2_Understand safety issues around vomiting before treating it

Antiemetics for motion sickness

Treating motion sickness

Since there are four completely different pathways to the vomiting centre, using four distinctly different sets of neurotransmitters, emesis (vomiting) needs to be treated with the correct antiemetic. This lesson helps you understand which antiemetics must be used to address motion sickness and which ones should be used for the type of vomiting that originates from the higher brain centre (like the vomiting that occurs when you think "looking at that just made me sick"').

Video: Antiemetics for motion sickness

https://www.youtube.com/watch?v=BUPW_TbI2dw

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_5_3_Antiemetics for Motion Sickness

Antiemetics for chemotherapy and GIT origins

Video: Antiemetics for chemotherapy and GIT

https://www.youtube.com/watch?v=AH68S5O-gqs

origins

Given that there are four distinctly different pathways to the vomiting centre, the correct medication is determined by the cause of the vomiting. This lesson explains which is most appropriate by using a clinical scenario.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_1_5_4_Antiemetics for Chemotherapy and GIT Origins

Activity: Watch this summary video

Module summary

Congratulations on completing the most difficult topics in Pharmacology!

Please reward yourself for giving this module your best shot. You will now understand the material in subsequent chapters, and most of the discussions about medications that you will engage in with colleagues, and you will be able to explain medications to clients. You have also begun your preparation for the 30% end of Term quiz.

See you in Module 2!

Warm regards, Ken

Video: Summary of chapter

https://www.youtube.com/watch?v=kam6obU6aGk

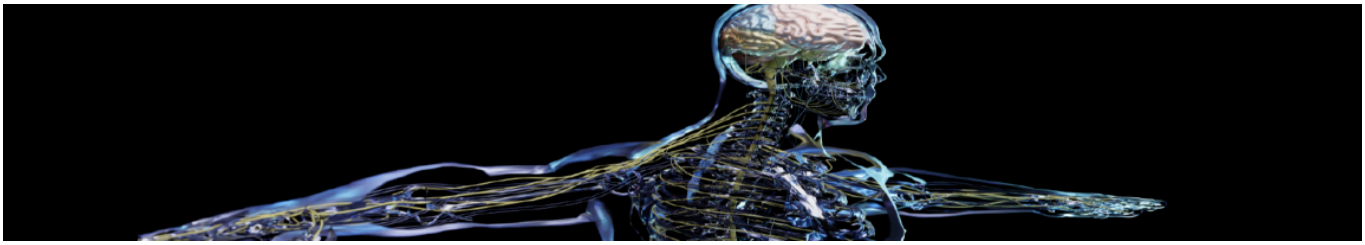
5

Module 2: The Respiratory and Autonomic Nervous Systems



The autonomic nervous system is the portion of the nervous system that regulates our most important involuntary physiologic processes. Module 2 allows you to picture the drugs that work by affecting the Autonomic Nervous System (ANS) and the drugs that treat disorders of the Respiratory System.

Module overview



Module overview: **Autonomic Nervous System (ANS) and the Respiratory system**

In this module, we explore the medications of the Autonomic Nervous System (ANS) and the Respiratory system. These are very important topics.

This entire module contains a total of 2 hours 15 minutes of video lessons plus activities that reinforce your knowledge.

Chapter 2.1: Foundational information for understanding Autonomic Nervous System (ANS) Pharmacology

Activity: Watch this video introduction to Chapter 1

Video: Introduction to chapter 1

YouTube link.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_1_0_Introduction_to_Chapter

Objectives, PowerPoint and Notes

The objectives for this chapter include:

- Describe the general properties of the Autonomic nervous system (ANS) (including the bodily functions it helps regulate, its two neuron nature, the neurotransmitters used, and the enzymes that degrade those neurotransmitters in the synaptic cleft).
- Give the other names of a sympathomimetic drug (for instance, adrenergic agonist), a sympatholytic drug (for instance, an adrenergic antagonist or blocker), an acetylcholine agonist (for instance, a muscarinic drug) and an acetylcholine antagonist (for instance, an antimuscarinic drug)
- Name the structures affected by the alpha 1, the beta 1 and the beta 2 receptors of the sympathetic nervous system, and for each of those structures note the effect of a 1. Sympathomimetic drug, and 2. Sympatholytic drug.
- Compare the actions of the sympathetic and parasympathetic nervous systems on the heart, the pupils of the eyes, the entire gastrointestinal system including the amount of saliva that is produced, the lungs, and the urinary bladder.

The PowerPoint and Notes for this chapter are linked directly below.

Overview of the Autonomic Nervous System (ANS)

Introduction to the nervous system

The various divisions of the nervous system can be very confusing. Worse yet, if the information is not fresh in your mind, it is impossible to understand the rest of material in this module. So let's engage in this quick review that places the autonomic nervous system (ANS) into the context of the other portions of the nervous system.

Video: Overview of Autonomic Nervous System

Watch this video, and perform the two drag and drop exercises below. Remember that the questions in the 30% quiz are derived from these activities so if you don't score well on your first attempt, try the activity again!

ADJUSTING SPEED OF VIDEO: Remember that you can listen to the reviews at a faster speed if you feel you already know the information. **It is a slightly different process to the videos from the last chapter,** where I put the material on YouTube. After the video starts, there is a Playback rate adjustment at the bottom centre portion of the video, that can be set at 0.5x, 1x, 1.4x. 1.6x and 2x. You can watch the video in YouTube right here.

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PHAR2001_2_1_1_Overview_of_the_ANS

Activity: Test your knowledge

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PHAR2001_2_1_1_Overview_of_the_ANS

Divisions of the ANS

The two divisions of the ANS

The two divisions of the autonomic nervous system are the Sympathetic Nervous System (SNS) and the Parasympathetic Nervous System (PNS). The SNS and PNS have, for the most part, opposite effects because the autonomic nervous system helps maintain homeostasis. So, if our heart was beating too slow for the conditions at the time, our cells would start getting too little oxygen, lactic acid would start to accumulate and homeostasis would be disrupted. But that does not happen because the SNS comes to the rescue, sending out a bit of adrenaline and homeostasis is maintained.

Video: Divisions of the ANS

After watching this video and you should be able to answer the multiple choice question below. **These activities will make sure that you can apply logic to understanding the actions of the SNS and the PNS to our various organs.** You can view the video on YouTube right here.

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PHAR2001_2_1_2_Divisions_of_the_ANS

Activity: Test your knowledge

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PHAR2001_2_1_2_Divisions_of_the_ANS

Overviewing drugs of the ANS

Overview

In this lesson, we outline one simple rule (along with two exceptions to the rule) that allows you to easily understand the drugs of the ANS.

By watching this video and doing the associated activities, you will streamline your work involved in understanding of the drugs of the ANS.

Just a few terms that are extremely important at this point:

- Sympathomimetics are drugs that mimic the sympathetic nervous system, like adrenaline and dopamine. They can also be called sympathetic nervous system agonists. While they often have a similar clinical outcome to the ParasympathoLYTIC drugs, THEY ARE NOT.
- Sympatholytic drugs are those that BLOCK the sympathetic nervous system, like the beta blockers. They can also be called sympathetic nervous system antagonists. While they often have a similar clinical outcome to the ParasympathoMIMETIC drugs, THEY ARE NOT.
- Parasympathomimetics are drugs that mimic the parasympathetic nervous system (the rest and digest system), like acetylcholine, although acetylcholine itself cannot be used as a drug because of its incredibly short half life. They can also be called parasympathetic nervous system agonists.
- A parasympathoLYTIC drug is a drug that is against the parasympathetic nervous system. It blocks acetylcholine and is often also called an anticholinergic drug or an antimuscarinic drug.

Video: Drugs of the ANS

View this video on YouTube here.

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PHAR2001_2_1_3_Overviewing_drugs_of_the_ANS

Activity: Test your knowledge

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PHAR2001_2_1_3_Overviewing_drugs_of_the_ANS

Selective receptors of the ANS

Understanding receptors of the ANS

The concept of subtypes of the alpha1, 2, 3, and 4-adrenoceptor was first suggested in the mid 1980s and since then we have started to develop more and more selective medications. Yes, it made our work as students a bit more difficult but it really decreased the side effects from such medications. Now, we can affect the heart without affecting the salivation. We can affect the peripheral blood vessels without affecting the lungs.

It takes a bit of practice but you will start to understand the side effects of a medication simply by knowing which receptors it binds to after engaging with this lesson.

Video: Selective SNS receptors

After watching this video, you should be successful in the three drag and drop exercises below. View this video on YouTube here.

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PHAR2001_2_1_4_Selective_receptors_of_the_ANS

Activity: Test your knowledge

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Clone of PHAR2001_2_1_4_Selective_receptors_of_the_ANS

Looking to the future

Video: Conclusion of chapter 1

Thanks for participating in these activities. As you will see, it would be nearly impossible to understand the medications of the Autonomic Nervous System without having this overview in your mind. Right now we are going to move on to the medications of the ANS, dividing them into SNS agonists, SNS antagonists, PNS agonists, and PNS antagonists.

View this video on YouTube here.

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PHAR2001_2_1_5_Chapter_1_Signing_Off

Chapter 2.2: Sympathomimetics (drugs that mimic the sympathetic nervous system)

Activity: Watch this video introduction to the chapter

Video: Introduction to chapter 2

View this video on YouTube.

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PHAR2001_2_2_0_Introduction_to_Chapter_2_of_the_ANS

Objectives, PowerPoint and Notes

The objectives for this Chapter include:

Describe the actions, interactions and side effects of the following adrenergic agonists:

- Adrenaline (non-selective agonist)
- Phenylephrine (an alpha 1 selective agonist)
- Dobutamine (a beta 1 adrenergic agonist)
- Salbutamol (a beta 2 adrenergic agonist)
- Dopamine

These medications will be taken again at a later time. It is most important right now to understand why the actions and side effects occur

The PowerPoint and Notes for this Chapter are directly below:

Anaphylaxis and adrenaline

Introduction to non-selective sympathomimetics

In this lesson, we begin our discussion of the non-selective **sympathomimetics** . Long name - what does that mean? Well, a sympathomimetic means that it is a drug that acts like adrenaline. It **mimics** the **sympathetic** nervous system.

Specifically in this lesson, we will get a good picture of what happens in anaphylactic shock, and why adrenaline is the main therapy and why antihistamines can also be of value.

Video: Anaphylaxis and adrenaline

Please pay close attention in this video - this is such an important topic that I have included 5 quick

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_2_1_Anaphylaxis_and_Adrenaline

activities that reinforce your knowledge! View this on YouTube.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_2_1_Anaphylaxis_and_Adrenaline

Other non-selective adrenergic agonists

Adrenaline and pseudoephedrine

Aside from adrenaline itself, the non-selective adrenergic agonists also include pseudoephedrine, which is a very effective decongestant and dopamine, which is pretty much only used in acute heart failure where the client needs a lot of stimulation to the heart in order to survive. We briefly look at both of those in this lesson.

Video: Non selective adrenergic agonists

Watch this video and reinforce your knowledge with the multiple choice and T/F questions below. View this video on YouTube.

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PHAR2001_2_2_2_Other_non-selective_adrenergic_ agonists

Activity: Test your knowledge

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PHAR2001_2_2_2_Other_non-selective_adrenergic_ agonists

Selective alpha agonists

Introduction to selective alpha agonists

The alpha-adrenergic agents will bind to and act like adrenaline on the peripheral blood vessels, the radial muscles of the eyes that controls the diameter of the pupils, the smooth muscles of the GIT, the male prostate and the urinary sphincter. They act like adrenaline on those sites. So please **predict the actions of those agents before clicking onto this next video** to see if you are right.

Video: Selective alpha agonists

Please engage in the video and the two activities below. Remember that if things are ever going too slow or too fast for you, you can set the video play speed higher or lower. View this video on YouTube here.

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PHAR2001_2_2_3_Selective_alpha_agonists

Activity: Test your knowledge

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PHAR2001_2_2_3_Selective_alpha_agonists

Selective beta agonists

Questions

What would a Beta 1 agonist be used for? ...a Beta 2 agonist?

Remember Beta 1 is the heart

Beta 2 is on the lungs and the mast cells. **Please attempt this before you click onto this video lesson** .

Video: Selective beta agonists

Please watch the video and then answer the questions to make sure you actively know the material. View this video on YouTube here.

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PHAR2001_2_2_4_Selective_beta_agonists

Activity: Test your knowledge

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PHAR2001_2_2_4_Selective_beta_agonists

Summary of Chapter 2

Video: Summary of chapter 2

Please join me as I rip through a quick 2 minute summary of this

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chapter. View this video on YouTube.

PHAR2001_2_2_5_Summary_of_Chapter_2

Chapter 2.3: Sympatholytic Drugs (drugs that block the sympathetic nervous system)

Activity: Watch this video introduction to the chapter

Video: Introduction to chapter 3

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_3_0_Introduction_to_Chapter_3

Objectives, PowerPoint and Notes

The objectives for this chapter includes to describe the actions interactions and side effects of the following adrenergic **antagonists** (blockers):

- Phentolamine and labetalol (non-selective antagonist)
- Prazosin and doxazosin (an alpha 1 selective antagonist)
- Metoprolol and atenolol (a beta 1 selective adrenergic antagonist)

Please feel free to use the PowerPoint and Notes linked directly below.

Understanding sympatholytics

Introduction to sympatholytic drugs

Sympatholytic drugs are drugs that block the sympathetic nervous system. Other names for the drugs include sympathetic nervous system antagonists, adrenergic antagonists, adrenergic blocking agents, there are many selective agents like alpha-blockers and beta-

blockers. These are NOT the same as a parasympathomimetic agent but they do allow the parasympathetic rest and digest system to DOMINATE. **So, predict what would happen with a sympatholytic drug** .

Video: Sympatholytic agents

Please engage in these activities that will help you understand subsequent material about drugs that act against the sympathetic nervous system. Watch this on YouTube here.

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PHAR2001_2_3_1_Understanding_sympatholytics

Activity: Test your knowledge

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PHAR2001_2_3_1_Understanding_sympatholytics

Non-selective adrenergic blockers

Video: Non-selective adrenergic blockers

There are only two non-selective adrenergic blockers that we need to remember, labetalol and phentolamine, but they are quite important medications.

Watch this quick video and answer the two associated multiple choice questions below. **Remember that**

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PHAR2001_2_3_2_Non-selective_adrenergic_blockers

the questions in the Week 6 quiz (worth a total of 30% of your grade in the unit) are directly from the material in these activities and you need to score 12 out of 30 possible points in order to pass the unit. (not necessarily the same exact questions but the same material) These questions are designed to make sure you do really well in that quiz, and as such, work safely with medications in your future career . Watch this video on YouTube.

Activity: Test your knowledge

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PHAR2001_2_2_3_Selective_alpha_agonists

Alpha 1 blockers

Overview of the alpha 1 receptor

The alpha 1 receptor is found on peripheral blood vessels and the prostate. If you think about it, you will probably guess that they would be potentially useful in benign prostatic hyperplasia and hypertension. We talk about these in a couple spots in the unit, so this lesson is less than 2 minutes long.

Video: Aplha 1 antagonists

Just hit the play symbol. You know you want to :)

or watch this on YouTube.

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PHAR2001_2_3_3_Alpha_1_blockers

Activity: Test your knowledge

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PHAR2001_2_3_3_Alpha_1_blockers

Beta blockers

Overview of beta blockers

Beta blockers had been the primary therapy for hypertension, but ACE inhibitors, angiotensin II receptor antagonists, calcium-channel blockers, and thiazide diuretics are generally preferred over beta blockers for the treatment of hypertension in the absence of co-morbidities. Nonetheless, beta blockers are still an important medication for hypertension, especially if it involves **congestive heart failure** , the type of heart failure that is chronic (long term) and can be treated for 20 years or longer.

Video: Beta 1 blockers

There are four quick activities associated with this very important group of medications that end in the suffix ...olol. After watching the

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PHAR2001_2_3_4_ Beta-blockers

video, please attempt all four .
Watch this video on YouTube.

Activity: Test your knowledge

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PHAR2001_2_3_4_ Beta-blockers

Summary of Chapter 3

Video: Summary of chapter 3

You are just two minutes away from finishing this chapter. ...or just one minute away if you speed up the tape to 2X!. Watch this video on Youtube.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_3_5_Summary_of_Chapter_3

Chapter 2.4: Pharmacology of the Parasympathetic Nervous System (PNS)

Activity: Watch this video introduction to Chapter 4

Video: Introduction to chapter 4

YouTube link.

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PHAR2001_2_4_0_Introduction_to_Chapter_4

Objectives, PowerPoint and Notes

The objectives for Chapter 4 include:

- Explain the difference between direct-acting and indirect-acting Parasympathomimetic drugs
- Know terms used to describe the parasympathomimetic drugs, for instance, muscarinic drugs, cholinergic agents, acetylcholine agonists.
- Describe the actions interactions and side effects of the:
  - Direct acting cholinergic agonists bethanechol and pilocarpine
  - Indirect acting cholinergic agonists neostigmine and donepezil
  - The antimuscarinic agents, using atropine as an example.

The PowerPoint and Notes for this chapter are directly below:

Understanding acetylcholine

Learning about acetlycholine

Recall that the neurotransmitter that is used at the neuromuscular junction that results in a contraction of the muscle is acetylcholine. And now we are saying that the neurotransmitter that works at the effector (at each of the organs of the ANS) is acetylcholine.

- So how is it that we can have medications that affect our organs without affecting muscle movement?

In this lesson, you will learn the answer to that question, and also learn the different names used for Parasympathetic Nervous System (PNS) drugs, and what their general effect will be if they cross into the brain.

Video: Understanding

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acetylcholine

You have to be curious about these beautiful mushrooms.

In the next lesson, you find out what will happen if you served these mushrooms up to your dinner guests. ...but you won't fully understand unless you watch this video! So click onto the play button and then engage in the four activities below.

YouTube link.

viewed online inside Blackboard.

PHAR2001_2_4_1_Understanding_acetylcholine

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_4_1_Understanding_acetylcholine

Understanding the PNS

Video: The parasympathetic nervous system

Take a deep breath, relax and think about what THAT is going to do to your organs...

You find out all about those pretty mushrooms in this lesson. Make

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PHAR2001_2_4_2_Understanding_the_Parasympathetic Nervous System (PNS)

sure you engage in all three activities after watching the lesson.

YouTube video link.

Activity: Test your knowledge

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PHAR2001_2_4_2_Understanding_the_PNS

Direct vs indirect cholinergic agents

Differences between direct and indirect cholinergic agents

For the most part, our ANS medications are receptor agonists and antagonists. But one notable exception is the indirect acting cholinergic agents that are enzymatic inhibitors. They act on and inhibit the enzyme that degrades acetylcholine. Let's take a look at the differences between direct and indirect cholinergic agents.

Video: Direct vs indirect cholinergic agents

Please watch this video and engage in the drag the word activity below.

YouTube video link.

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PHAR2001_2_4_3_Direct_vs_indirect_cholinergics

Activity: Test your knowledge

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PHAR2001_2_4_3_Direct_vs_indirect_cholinergic_agents

Indirect acting acetylcholine agonists

Understanding Indirect acting acetylcholine agonists

The most common type of nerve gas irreversibly inhibits the enzyme acetylcholinesterase. Think about what that would do. There would be huge amounts of acetylcholine in the synaptic cleft because it is not degrading. But where would that excess be? In the MUSCLES, meaning there would be constant contraction - how can you breath if the diaphram is not able to relax? In the HEART, meaning there would be a profound bradycardia (decreased heart rate). In the GIT, giving excess salivation, cramping and diarrhoea. Now lets back up a bit and think about when a reversible version of that could be useful.

Video: Indirect acting acetylcholine agonists

Engage in this video and multiple choice question to understand the two different kinds of acetylcholine agonists.

YouTube video link.

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PHAR2001_2_4_4_Indirect_acting_acetylcholine_agents

Activity: Test your knowledge

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PHAR2001_2_4_3_Direct_vs_indirect_cholinergic_agents

Anticholinergic agents

Video: Anticholinergic agents

You have probably heard someone say that their medication gives them a dry mouth. Why is it that so many of our medications give the person taking them a dry mouth?

Find out more about this pretty plant by watching the video and engaging in the two activities below.

YouTube video link.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_4_5_Anticholinergic_agents

Activity: Test your knowledge

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PHAR2001_2_4_5_Anticholinergic_agents

Summary of Chapter 4

Video: Summary of chapter 4

YouTube video link

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Sorry, I did say ' Ya' in the video. ... and I did imply that this was the last Chapter in this module, which it is not.

One more chapter to go, because I decided to put the Respiratory system pharmacology AFTER the ANS. But most of our important drugs of the respiratory system are based on drugs of the ANS, so you are well on your way to finishing the Module!!

PHAR2001_2_4_6_Summary_of_Chapter_4

Chapter 2.5: Pharmacology of the Respiratory System

Introduction to Respiratory System Pharmacology

Video: Introduction to chapter 5

In this final chapter of Module 2, we explore the respiratory system pharmacology. Our objectives for this chapter include the following:

- Briefly describe the structure and function of the lungs including the immune system response in the lungs
- Briefly describe the pathophysiology of allergic rhinitis and asthma identifying and describing the actions of common pharmacological agents used in their treatment

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_5_0_Introduction_to_Respiratory_System

- Compare the difference between mucolytic agents and expectorants, citing examples of each.

YouTube video link.

The PowerPoint and Notes for this chapter are directly below:

Preview of respiratory system

Review of the respiratory system

This lesson reviews the important points about the respiratory system that will help you understand the relevant medications.

The respiratory system is broadly divided into the upper and lower respiratory system. Upper respiratory system disorders that are treatable with medications include allergic rhinitis and the common cold, and our main drugs used to treat these include the antihistamines and the decongestants.

The lower respiratory system disorders that are treatable with our medications include those disorders that result in obstruction of the airways, which are sometimes referred to as the ventilation disorders. Common medications used to treat those disorders include bronchodilators, inhaled corticosteroids and oral corticosteroids. If the disorder results in thick or copious amounts of mucus, the expectorants or mucolytic agents may also be of value. We also have antitussives or cough medicines that work at the level of the medulla to decrease the urge to cough, but those should only be used for non-productive coughs.

There are several other disorders that involve the respiratory system, including pleural effusions, and pulmonary embolism, however those disorders are covered elsewhere, according to the origin of the complaint.

Video: Respiratory system review

This is an applied review, that includes the information about respiratory system disorders that you need to know in order to understand the drugs of the respiratory system. If you think you already know enough about the respiratory system that a 'review' will not be needed, watch the video anyway, and turn it up to 2X speed. (After the video starts, click onto the 1X symbol a couple times till it reads 2X, which is quite fast.)

YouTube video link

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_5_1_Review_of_the_Respiratory_System

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_5_1_Introduction_to_Respiratory_System_Pharmacology

Antitussives expectorants and mucolytic agents

Overview of antitussives, expectorants and mucolytic agents

Although they often do not require a prescription, antitussives, expectorants and mucolytic agents are all widely used, and as such, very important to understand.

Antitussives (or cough suppressants) that are correctly named will work at the level of the medulla, and include the opioids and dextromethorphan. Other medications that soothe the lungs, or get mucus off the cough receptors are not termed antitussives.

Expectorants and mucolytic agents generally aim to do the same thing, to get mucus out of the lungs, but they work by different mechanisms. In this lesson, we explore these three important classes of medications in much more depth.

Video: Antitussives, expectorants and mucolytic agents

Most of the drugs discussed in this lesson are over the counter, but SOOO important to understand for everyone from a pharmacy assistant to an MD. Watch the video and then test your knowledge with the other activities.

YouTube video link.

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PHAR2001_2_5_2_Antitussives_expectorants_mucolytics

Activity: Test your knowledge

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PHAR2001_2_5_2_ Antitussives_ expectorants_and_mucolytic_agents

Decongestants and antihistamines

The role of decongestants and antihistamines

When an immune response happens in the sinuses, histamine is released from the local mast cells, and the histamine dilates the blood vessels and increases the permeability of those blood vessels. The escaping fluid and cells work on the mucus secreting cells, resulting in both plasma and mucus being excreted. An antihistamine blocks the action of histamine on the receptors and a decongestant does the opposite thing that histamine does, contracting the same blood vessels that histamine dilated.

In this lesson, we concentrate on understanding that process and on the decongestants, as our tutorial for this week fully covers the antihistamines.

Video: Decongestants and antihistamines

Watch this video and engage in the M/C and T/F questions.

YouTube video link.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_5_3_Decongestants_Antihistamines

Activity: Test your knowledge

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PHAR2001_2_5_3_ Decongestants_and_Antihistamines

Lower respiratory system disorders

Overview of the lower respiratory system

The lower respiratory system disorders that are able to be treated by medications fall under the classification of the ventilation disorders, or in other words, the ones that obstruct the airways. Asthma is the most common of these, affecting over 10% of Australians. Since it is the most common and because of the fact that the medications used to treat asthma are often the ones that treat other lower respiratory system disorders, we aim to get a good picture of these in this lesson.

Video: Lower respiratory system disorders

Asthma is the most common lower respiratory system disorder in Australia. Learn the most important facts about the disorder by watching this video and engaging in the associated activities.

Youtube video link.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_5_4_Lower_Respiratory_System_Disorders

Activity: Test your knowledge

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PHAR2001_2_5_4_Overview_of_Asthma

Treatment of Asthma (and other Ventilation Disorders)

Pharmacological management of asthma

As you have learned in the previous lessons, when an person experiences an asthmatic attack, the mast cells in their lungs degranulate and release histamine. So you would think that antihistamines are going to be helpful for asthmatics, but only about 10% of asthmatics get any additional benefits from them, and they should never be relied on for an acute asthmatic reaction. The main drugs for asthma are the bronchodilators, and those can be sympathomimetics, parasympatholytics, or other substances that dilate the lungs. Another very important target of our medications will be drugs that target the actual mediators of inflammation. The leukotrienes are mediators of inflammation that are often targeted, but the most useful medication that targets the immune system will be the corticosteroids (either inhaled or systemic). Corticosteroids decrease all arms of the immune system, so they are very important in the treatment of asthma.

Video: Treatment of Asthma

Remember that although asthma is regarded by many authors as a reversible disorder, it does involve chronic inflammation, and chronic inflammation almost always results in fibrosis and scarring, remodelling of the tissues, and permanent changes, so asthma must be treated and you must know how it is treated because 30% of Australians will need these asthma meds during at least one period of their lives.

YouTube video link.

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PHAR2001_2_5_5_Treatment_of_Asthma

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_2_5_5_Treatment_of_Asthma_and_other_Ventilation_Disorders

Congratulations on finishing the final Chapter in Module 2!

Video: Module 2 summary

Just one minute and 14 seconds left in this module, so quick, click on that arrow before I add more material!

Youtube video link.

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PHAR2001_2_5_6_Signing_off_Module_2

Module 3: The Endocrine System



The endocrine system is a messenger system that regulates a range of bodily functions. This module explores that system, the disorders that can occur to that system and the medications which aim to treat endocrine system disorders.

Module overview



Module overview: The endocrine system

The endocrine system is essentially a communication system. It depends on feedback loops of the hormones released by internal glands directly into the circulatory system that regulate distant target organs. But what happens when that communication system breaks down? This module answers that question.

This entire module contains a total of 70 minutes of video lessons plus activities that reinforce your knowledge and prepare you for the quiz.

Chapter 3.1: Introduction to the Endocrine system, ADH, Oxytocin and GH

Introduction to Endocrine System

Video: Introduction to the endocrine system

Join me in this video that introduces you to the endocrine system.

Watch on YouTube here.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_3_1_0_INTRODUCTION_to_Endocrine_System

Chapter objectives

The objectives for this chapter include:

- Define the 'endocrine system' and 'hormone'
- Give an example of a homeostatic control mechanism involving 'negative feedback'.
- Explain the reasons that the hypothalamus is thought to be the controlling organ of the endocrine system.
- Explain the effects of the two hormones secreted from the posterior pituitary gland (oxytocin and ADH) and give an example of when they would be useful as a pharmaceutical agent.
- Explain the actions of growth hormones and the results of excesses and deficiencies during childhood and after puberty.
- Drugs to know: Desmopressin, somatropin, vasopressin, oxytocin

Please see the links below for the PowerPoint and Notes for the first two chapters of this module:

Review Part 1

A review of the endocrine system: Part 1

As we begin discussing the endocrine system, it is important that we have the overview of the system in our mind. The first two lessons review the endocrine system. Remember that you can speed up the videos, to 1.6X or 2X if you have taken two units of physiology recently.

Video: Review of the endocrine system

Watch this quick review of the endocrine system to refresh your memory of the clinically relevant information about the endocrine system.

Youtube video link.

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PHAR2001_3_1_1_Review_Part_1

Activity: Test your knowledge

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PHAR2001_3_1_1_Review_Part_1

Review Part 2

Video: Review of the endocrine system - Part 2

This lesson contains the final 6 minutes of the endocrine system review.

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PHAR2001_3_1_2_Review_Part_2

Activity: Watch this video

Youtube video link.

Activity: Test your knowledge

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1

Oxytocin

Video: Medications of the endocrine system: Oxytocin

Oxytocin, 'the love hormone'.

Find out why oxytocin is called the love hormone by watching this video, and then reinforce your knowledge by engaging in the activity below.

Youtube video link.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_3_1_4_Oxytocin

Activity: Test your knowledge

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PHAR2001_3_1_4_Oxytocin

ADH water saver (not compulsory for this unit)

Vasopressin - The **antidiuretic hormone**

The material in this lesson will not be assessed in PHAR2001.

Human vasopressin, also called antidiuretic hormone (ADH) is a hormone synthesized from the hypothalamus. It then travels down the axon terminating in the posterior pituitary, and is released into the circulation in response to hyperosmolality (equivalent to dehydration).

ADH is the water saver, and (its first metabolite) saves the water at its last chance of being reabsorbed at the level of the kidneys just before it is excreted into the urinary bladder. As you would expect, as ADH saves water, it has the effect of increasing the blood pressure.

Diabetes insipidus is a disorder characterised by a lack of ADH and while it is relatively uncommon, it is important to understand ADH to understand desmopressin (the treatment for diabetes insipidus) and to understand many drugs, because there are several that affect ADH. For instance, caffeine decreases ADH release and ethanol (drinking alcohol) decreases its production, both leading to their 'diuretic' effects. So let's find out more about this important hormone in this lesson.

Video: The **antidiuretic hormone** (Not compulsory for this unit)

Youtube video link.

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PHAR2001_3_1_3_ADH_water_saver

Activity: Test your knowledge

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PHAR2001_3_1_3_ADH_water_saver

Growth hormone (not compulsory for this unit)

Somatotropin - The growth hormone

The material in this lesson will not be assessed in PHAR2001.

Growth hormone or somatotropin, is a hormone that stimulates growth, cell reproduction, and cell regeneration and is vital in human development.

Video: Growth hormone excess

Youtube video link.

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PHAR2001_3_1_5_Growth_Hormone

Activity: Test your knowledge

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2

Sign off chapter 1

Video: Chapter 1 summary

Youtube video link.

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PHAR2001_3_1_6_sign_off_chapter_1

Chapter 3.2: The Thyroid Axis

Understanding thyroid hormone

Introduction to the thyroid hormone

If you could design a hormone that could be released when you are cold, what would it do? It would speed up the metabolism of nearly all cells in the body. The sodium-potassium ATPase pumps all through the body would start working overtime, the body would get much more sensitive to adrenaline.

Well, that is the thyroid hormone. It is also important in growth, metabolism and is absolutely necessary for human life. Let's learn more about the thyroid hormone by engaging in this lesson.

Video: The thyroid gland

Remember that you can adjust the speed of these videos to 0.5x, 1x, 1.4x, 1.6x and 2x. The slow speed is very difficult to understand, and as such, I speak a bit slower. As such, one of the faster speeds may be the correct speed for you. Once the video starts playing, simply adjust the 1X to the speed that suits your learning.

Sorry, in the video, I mentioned that the thyroid gland hypertrophies (cells get larger) when there is inadequate iodine, but that process involves more of a **hyperplasia** (increased number of cells) than a

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PHAR2001_3_2_1_Understanding_Thyroid_Hormone

hypertrophy, although both will be occurring. The notes are correct.

Youtube video link.

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_3_2_1_Understanding_Thyroid_Hormone

Hypothyroidism and its treatment

Video: Hypothyroidism

Hypothyroidism occurs in about 5% of all women and increases with age, although it is less common in men. Please take the time to learn more about this important disorder with this lesson.

Youtube video link.

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PHAR2001_3_2_2_Hypothyroidism_and_its_Treatment

Activity: Test your knowledge

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PHAR2001_3_2_2_Hypothyroidism_and_its_Treatment

Hyperthyroidism and its treatment (Not compulsory)

Treatment of hyperthyroidism

The polar opposite disorder to hypothyroidism is obviously hyperthyroidism. Interestingly, the most common type of this disorder is also an autoimmune disorder. But in this case the immunoglobulins produced will be acting like thyroid stimulating hormone, resulting in excessive release of T4. Let's take a closer look at this group of disorders and their treatment in this final lesson of the chapter.

Video: Hyperthyroidism symptoms (not compulsory)

Youtube video link.

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PHAR2001_3_2_3_Hyperthyroidism_and_its_Treatment

Activity: Test your knowledge

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PHAR2001_3_2_3_Hyperthyroidism_and_its_Treatment

Signing Off

Video: Summary of chapter 3

Hi All, In this final video, I mention that diabetes will be the next Chapter, but I decided to present Diabetes within the tutorial and the

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PHAR2001_3_2_4_Signing_Off

case studies, so you will have adequate coverage of that disorder within those activities. Regards, Ken

Youtube video link

Chapter 3.3: Diabetes and Metabolic Syndrome

Chapter 3.4: The Adrenal Axis

Introduction Stress Axis

Video: The adrenal axis

After watching this video introduction to the topic, you can download the notes directly below and go on to the rest of the chapter.

Youtube video link.

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PHAR2001_3_4_0_Introduction_Stress_Axis

Objectives

The objectives for this chapter include:

- Name and explain the actions of the two types of steroidal hormones released from the adrenal cortex (glucocorticoids and mineralocorticoids – the androgens are not as important so concentrate on the first two).
- Using hypoglycaemia as the stressor explain why the 'hypothalamus > anterior pituitary > adrenal cortex' axis is considered the stress axis (in other words, how does this axis help out in hypoglycaemia?)
- Using dehydration or hypovolemic shock as the stressor explain why the 'hypothalamus > anterior pituitary > adrenal cortex' axis is considered the stress axis (How does the adrenal gland help in times of dehydration?) Drugs to know: Spironolactone and Fludrocortisone

- Detail the physiological actions of the glucocorticoids. Give specific examples of the application of glucocorticoids in pharmacology, using specific drugs as examples (Methylprednisolone, Prednisolone and Dexamethasone). List ten specific adverse effects of glucocorticoids.

The notes and PowerPoint for this chapter are directly below:

Activity: Test your knowledge

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PHAR2001_3_4_0_Introduction_Stress_Axis

Overview Stress Axis

Overview of the corticosteroids

In this lesson, we will start understanding the stress access. When we are dying of dehydration or of hypoglycaemia, there is a cascade of hormones that ultimately results in the adrenal cortex secreting the stress hormones. Some of those are termed corticosteroids, and others include noradrenaline. The Corticosteroids include the GLUCOcorticoids which save glucose and the MINERALcorticoids that save salt. These hormones are so valuable in times of stress that they are the obvious choice to mimic in a pharmacological sense.

Please take time to understand this overview of the corticosteroids before getting into the lessons for this chapter.

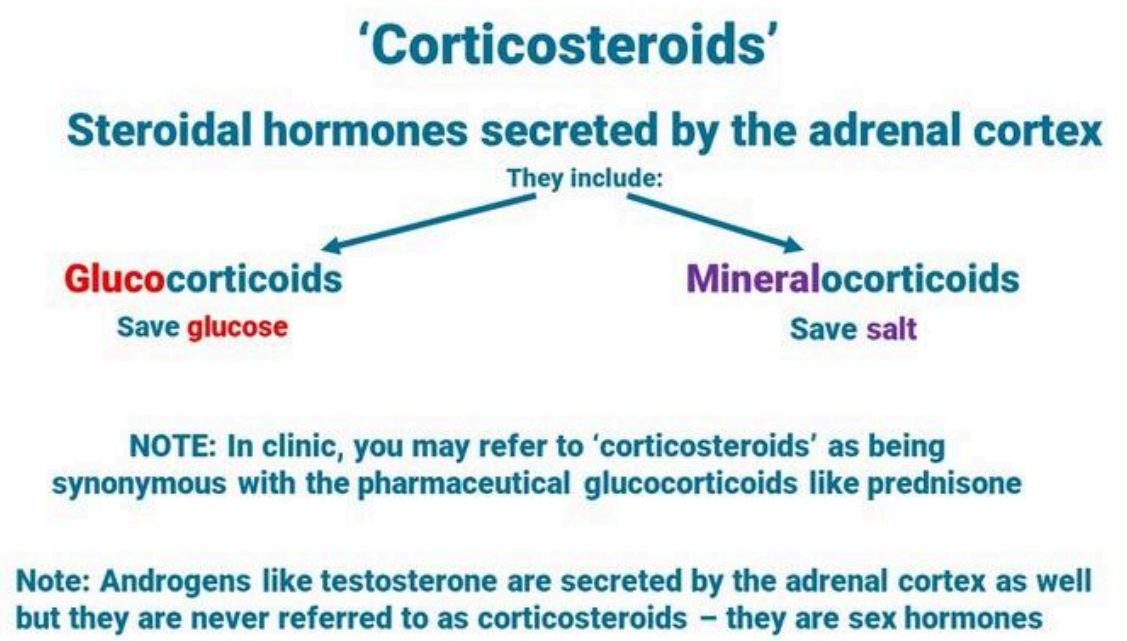


Figure 3.1 : Corticosteroids - The Steroidal hormones secreted by the adrenal cortex

Video: Stress response system

Youtube video link.

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PHAR2001_3_4_1_Overview_Stress_Axis

Activity: Test your knowledge

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PHAR2001_3_4_1_Overview_Stress_Axis

Glucocorticoids

Video: Glucocorticoids

In this lesson, we begin to really understand the side effects of long term 'corticosteroid' (glucocorticoids; drugs like prednisone) treatment by picturing the actions of cortisol.

Watch this video to really get a good picture of the actions of the glucocorticoids like cortisol.

Youtube video link.

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PHAR2001_3_4_2_Glucocorticoids

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_3_4_2_Glucocorticoids

Side effects of corticosteroids

Video: Side effects of corticosteroids

Watch this video to really picture the side effects of long term corticosteroid treatment.

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PHAR2001_3_4_3_Side_Effects_of_Corticosteroids

Youtube video link.

Activity: Test your knowledge

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PHAR2001_3_4_3_Side_Effects_of_Corticosteroids

Signing off

Video: Conclusion of chapter 4

Just one last 1.5 minute video to conclude this chapter.

Youtube video link.

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PHAR2001_3_4_4_Conclusion

Chapter 3.5: The Endocrine System and Bone Health

Introduction to medications for bone health

Video: Introduction to chapter 5

In this final portion of the endocrine system, we explore the endocrine regulation that ensures optimal bone health.

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PHAR2001_3_6_0_Introduction_bone_disorders

Youtube video link.

Objectives

The objectives for this chapter include:

- Explain the effects of parathyroid hormone and calcitonin and their role in bone health.
- Explain the process of bone formation and reformation
- Using two specific examples (bisphosphonates and teriparatide), explain how pharmaceutical agents can be beneficial for people with bone loss. Explain the main adverse drug reactions of these medications.

If you would like the notes or PowerPoint presentation for this chapter, please click on the link(s) below:

Understanding bone health

Video: The bone remodelling process

All through our life, there is a breakdown and reformation of bone. This lesson reviews that process and overviews the most common bone disorder, osteoporosis.

Youtube video link.

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PHAR2001_3_6_1_Understanding_bone_health

Activity: Test your knowledge

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PHAR2001_3_6_1_Understanding_bone_health

Understanding bone health 2

Video: Bone health

In this lesson, we complete our understanding of bone health.

Youtube video link.

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PHAR2001_3_6_2_Understanding_bone_health_2

Activity: Test your knowledge

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PHAR2001_3_6_2_Understanding_bone_health_2

Bisphosphonates Teriparatide SERMS

Video: Other treatments for osteoporosis

Youtube video link

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PHAR2001_3_6_3_Bisphosphonates_Teriparatide_SERMS

Activity: Test your knowledge

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PHAR2001_3_6_3_Bisphosphonates_Teriparatide_SERMS

Summary of Module 3

And that concludes the shortest Module of the unit. Please make sure that you have engaged in all of the activities, and then proceed to one of the average sized Modules in the unit, which has to do with the medications that treat cardiovascular diseases.

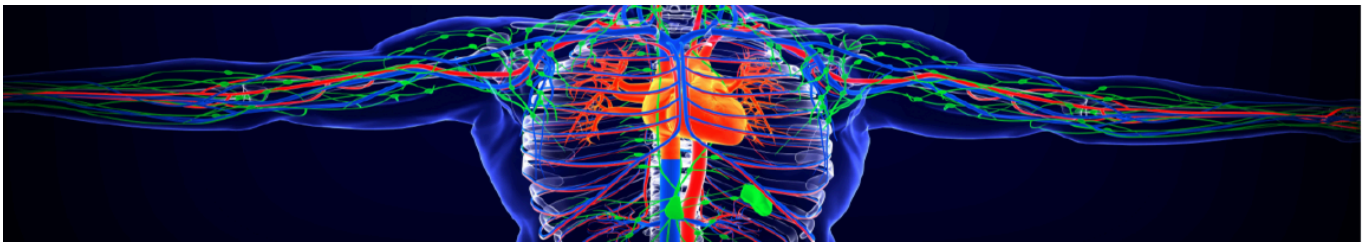
Module 4: The Cardiovascular and Renal System



An estimated 34% of Australians have hypertension and 27% of deaths in Australia are caused by cardiovascular disease. Learn about how we treat these individuals in this module.

Cardiovascular System Chapter 1: Lipids

Introduction to Module 4 Chapter 1



Module overview

This entire module contains a total of 2.5 hours of video lessons plus activities that reinforce your knowledge.

Introduction to chapter 1

Hyperlipidaemia is the silent killer. In this chapter, we learn about how they contribute to the disease process (atherosclerosis) that leads to myocardial infarction, stroke, and loss of limbs, as well as our medications that can help prevent its development.

The objectives for this chapter include"

- Define the terms hyperlipidaemia, dyslipidaemia, HDL and LDL
- Describe atherosclerosis, including the risk factors, and potential consequences of plaques on the arteries

- Identify and describe the therapeutic actions and major side effects of common pharmacological agents used in the treatment of dyslipidaemia or hyperlipidaemia.

The Notes and PowerPoint for this chapter can be found directly below:

Understanding Lipids

Introduction to lipids

Not all lipids are ‘bad’. They perform many functions both within the body, as they act as an energy reserve, regulate hormones, transmit nerve impulses, cushion vital organs, and transport fat-soluble nutrients. They are also important in the structure of cell membranes and hormones. This lesson helps you understand more about when it may be time to treat the lipid levels in the body.

Video: Understanding lipids

Join me in this video introduction to lipids.

Youtube video link.

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PHAR2001_4_1_1_Understanding_Lipids

Activity: Test your knowledge

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PHAR2001_4_1_1_Understanding_Lipids

Understanding atherosclerosis

Introduction to Atherosclerosis

'Atherosclerosis' is a strange word,what does it mean? Well, 'sclero' comes from the Greek sklērós, meaning “hard.” Atherosclerosis therefore literally means hardening of the arteries.

It is a chronic inflammatory condition that results in fibrosis, scarring and narrowing of the arteries. But more importantly, **atherosclerosis is the underlying cause of about 50% of all deaths in westernized society.** Therefore we should take the time to learn as much about the disorder as possible because many of our therapeutic agents will target the risk factors of the disorder that allow the problem to progress.

Video: Understanding atherosclerosis, Part 1

Youtube video link.

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PHAR2001_4_1_2_Understanding_Atherosclerosis

Activity: Test your knowledge

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PHAR2001_4_1_2_Understanding_Atherosclerosis

Understanding atherosclerosis, Part 2

Video: Understanding atherosclerosis, Part 2

In this lesson, we continue our understanding of the process of

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atherosclerosis.

Youtube video link.

PHAR2001_4_1_3_Understanding_Atherosclerosis_Part_2

Activity: Test your knowledge

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PHAR2001_4_1_3_Understanding_Atherosclerosis_2

The statins

Video: The statins

Given the importance of lipids in the process of atherosclerosis, we have several types of medications that aim to normalise the lipid levels. In this lesson, we explore the Hmg CoA reductase inhibitors, or more commonly referred to as the statins because all of the members of that drug class end in the suffix ...STATIN.

Youtube video link.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_4_1_4_The_Statins

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_4_1_5_Statins

Lowering cholesterol

Video: Cholesterol absorption inhibitors

Another way to decrease the cholesterol in the system is to prevent the uptake of cholesterol from the diet. It isn't the best method as a monotherapy but these are the only lipid lowering drugs that can be combined with others.

Youtube video link.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_4_1_5_Cholesterol_Absorption_Inhibitors

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_4_1_6_Cholesterol_absorption_inhibitors

Nicotinic acid

Video: Nicotinic acid

If my ears are burning, does that mean that someone is talking about me?

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Youtube video link.

PHAR2001_4_1_6_Nicotinic_Acid

Activity: Test your knowledge

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PHAR2001_4_1_6_Nicotinic_Acid

HDL levels

Dyslipidaemia and HDL

Recall that we use the term dyslipidaemia for individuals who have low levels of HDL, the good protein, because that is a risk factor for developing atherosclerosis. HDL is the 'bus' that carries cholesterol from the tissues back to the liver, so a low amount of the HDL will yield high amounts near the blood vessels. Nicotinic acid raises the HDL levels, but if that is not able to be tolerated, fenofibrate or other fibrates can be helpful. It is one of the few times that you will still see those medications still used.

Video: Fibrates

Youtube video link.

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PHAR2001_4_1_7_Fibrates

Activity: Test your knowledge

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PHAR2001_4_1_7_Fibrates

Summary

Video: Summary of chapter 1

We conclude this chapter by reiterating some important points about all lipid-lowering agents.

After watching this video, please check to make sure that you engaged in all of the quizzes and activities above, and move on to the next chapter where we will explore another risk factor for hypertension, which is hypertension. Remember that is the best preparation for the End of Term Quiz.

Youtube video link

Thanks for your participation in this chapter,

Regards, Ken

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PHAR2001_4_1_8_Summary_of_Lipids

Cardiovascular System Chapter 2: Hypertension

Introduction to Module 4, Chapter 2

Introduction to chapter 2: Hypertension

Nearly 25% of Australians suffer from hypertension, which is a risk factor for numerous chronic disorders. In this chapter, we will explore hypertension, beginning with an understanding of blood pressure regulation, and placing our most important antihypertensive agents into the context of that information. The understanding of blood pressure control is very important in our understanding of antihypertensive drugs because our medications target those physiological mechanisms.

The objectives for this chapter include:

- Explain the regulation of blood pressure, including the short-term regulation and the long term regulation.
- Identify and describe the therapeutic actions and major side effects of common pharmacological agents used in the treatment of hypertension

Please note that the Notes and PowerPoints for this chapter are directly below:

Understanding blood pressure control, Part 1

Regulating blood pressure

If we understand blood pressure regulation, we can understand our antihypertensive medications because everyone of our antihypertensive medications affect one or more of those mechanisms. So the next three short lessons explore blood pressure regulation.



Video: Blood pressure control, Part 1

Please watch these next three short videos to make sure that you understand some of the most frequently used medications in

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_4_2_1_Understanding_Blood_Pressure_Control_Part_1

Australia, the antihypertensive drugs.

Youtube video link

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_4_2_1_Understanding_Blood_Pressure_Control_Part_1

Understanding blood pressure control, Part 2

Video: Understanding blood pressure control, Part 2

If you understand the physiological mechanisms of blood control in the body naturally, you will understand the medications that control blood pressure. This lesson gives you that understanding.

Youtube video link.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_4_2_3_Understanding_Blood_Pressure_Control_Part_2

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_4_2_1_Understanding_Blood_Pressure_Control_Part_2

Understanding blood pressure control, Part 3

Video: Understanding blood pressure control, Part 3

This lesson concludes our understanding of blood pressure regulation, and our foundational information that is needed for us to understand our blood pressure medications.

Youtube video link.

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_4_2_3_Understanding_Blood_Pressure_Control_Part_3

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_4_2_3_Understanding_Blood_Pressure_Control_Part_3

Hypertension

Video: Complication of hypertension

Now that you have a good understanding of how we regulate blood pressure in a physiological sense, you will understand how our antihypertensive medications work. So in this lesson, we will begin our

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PHAR2001_4_2_4_Overview_of_Hypertension

discussion of antihypertensive agents by getting a good picture of hypertension itself.

Youtube video link.

Activity: Test your knowledge

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PHAR2001_4_2_4_Hypertension_Definition

The A, B, C, D's of the drug classes that are useful in hypertension

Hypertension Pharmacology made simple: The ABCD framework

When we're thinking about the drugs that are used to treat hypertension, one of the ways in which we can easily remember those drugs is to simply remember the first four letters of the alphabet. We have spoken about a few of these medications already, and the next few lessons will bring it all together for you. Remember, A, B, C and D:

- **A**lpha 1 blockers (...azosin drugs)
- **A**lpha 2 agonists (yes, agonists,
- **A**ngiotensin converting enzyme (ACE) inhibitors
- **A**ngiotensin 2 receptor blockers (ARBS; ...sartan)
- **B**eta blockers
- **C**alcium channel blockers (Dihydropyridines)
- **D**iuretics – Especially Hydrochlorothiazide or thiazide diuretics

So, with respect to the alpha blockers, they are adrenergic antagonists that selectively affect the peripheral blood vessels.

The drugs of the renin cascade, being the angiotensin converting enzyme (ace inhibitors) and angiotensin 2 receptor blockers (ARBS),

The beta blockers that block the adrenaline binding to the heart

The calcium channel blockers that do not allow the calcium into the cells of the heart (calcium is needed for contraction)

And the diuretics, which we will discuss in the last lesson of this module.

Let's move on to the first of those in this lesson.

Alpha Antagonists

Video: Alpha antagonists

The alpha 1 antagonists like prazosin were first line therapies for hypertension and also for benign prostatic hyperplasia. They have been largely superseded by medications that are more effective and safer, but they still have their place, normally with those who have both conditions simultaneously.

Youtube video link.

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PHAR2001_4_2_5_Alpha_1_Blockers

Activity: Test your knowledge

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PHAR2001_4_2_5_Alpha_Antagonists

Medications that affect angiotensin II

Targeting the RAA mechanism in hypertension

The renin-angiotensin-aldosterone (RAA) mechanism is our 'one-stop-shop' for increasing our blood pressure. By simply releasing one enzyme, we can create a cascade of events that ultimately result in the blood pressure increasing in five different ways.

Wouldn't it be a great antihypertensive medication if we could design a drug that stopped that cascade? Well we did - twice ! This lesson describes two drug classes that ultimately decrease the effects of the RAA mechanism, and those medications are both 'first line' therapies for hypertension.

Video: ACE inhibitors and ARBs

Youtube video link.

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PHAR2001_4_2_6_ACE_Inhibitors_and_ARBs

Activity: Test your knowledge

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PHAR2001_4_2_5_Renin drugs

Alternative drugs for hypertension

Why beta blockers are no longer first-line therapy for hypertension

Have you ever wondered how long it takes between the time that scientists discover that a drug class is no longer the first-line therapy for a certain disorder and the time that doctors stop prescribing it for that disorder? Well, it has been over ten years since better alternatives for hypertension have been found, and atenolol and other beta blockers are still the first choices for many doctors.

Lesson learnt: Try to keep up with the most recent information about the first-line therapies, because they do change, and quite frequently. This lesson explains why the beta blockers are no longer #1.

Video: Beta blockers

Youtube video link.

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PHAR2001_4_2_7_Beta_Blockers.

Activity: Test your knowledge

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PHAR2001_4_2_7_beta_blockers

Calcium channel blockers

Video: Calcium channel blockers

The muscles in blood vessels and the heart need calcium inside the

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muscle in order to contract, so decreasing the amount of calcium as it travels into the muscle sounds like a good mechanism to decrease blood pressure. Let's see how well the calcium channel blockers work.

PHAR2001_4_2_8_Calcium_Channel_Blockers

Activity: Test your knowledge

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PHAR2001_4_2_8_Calcium_Channel_Blockers

Final words about antihypertensive agents

Video: Final words about treating hypertension

Let's put it all together in this final lesson of the chapter.

Youtube video link.

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PHAR2001_4_2_9_Summary_of_Hypertension

Activity: Test your knowledge

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PHAR2001_4_2_9_Final words about antihypertensive agents

Cardiovascular System Chapter 3: Thrombi

Introduction to Module 4 Chapter 3

Objectives

The objectives for this chapter include:

- **Explain the various steps involved in thrombus formation**
- **Explain how each of our medications that decrease thrombus formation affect one of the steps involved in thrombus formation,**
- **Identify and explain the actions of common pharmacological agents used to prevent the development of a thrombus.**

You can find the notes and PowerPoint for this chapter directly below:

Video: Introduction to chapter 3 - Thrombosis

Please watch this video introduction to the chapter on thrombosis, or the formation of a blood clot.

Youtube video link.

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PHAR2001_4_3_4_INTRO_THROMBOSIS

Arterial clots

From haemostasis to thrombosis: mechanisms and consequences

Haemostasis is the process of stopping bleeding. We use it every day to repair small bleeds, and it is vital to our survival. But our lifestyle has lead to us being predisposed to this process occurring in absence of a break in the blood vessel walls. In that case, a thrombus develops. Thrombi can occlude the site in which it develops or they can break off from the original site

of formation to a blood vessel downstream. In order to combat the development of thrombi, we target the same process in which they develop, the platelet plug, the clotting cascade and the natural dissolving of the clot. Engage in this lesson to understand why our medications are effective.

Video: Thrombosis overview

Youtube video link.

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PHAR2001_4_3_5_THROMBOSIS_OVERVIEW

Activity: Test your knowledge

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PHAR2001_4_3_1_Overview_of_antithrombotic_agents

Antiplatelet drugs

The role of platelets in haemostasis and drug targeting

Platelets are invaluable in the process of haemostasis, and as such, most of our antiplatelet drugs will not disable the platelet altogether, but rather just take away one of the platelet's 'tools'. In order to understand our antiplatelet drugs, the most important thing to remember is that platelets are not cells, and as such, they do not contain the DNA to synthesise more 'tools'. So when we take away one of the platelet's tools that allows for optimal aggregation, that 'tool' will be disabled for the entire life of the platelet, which is 9 days.

Video: Antiplatelet drugs

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Youtube video link.

PHAR2001_4_3_6_ANTIPLATELET_DRUGS

Activity: Test your knowledge

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PHAR2001_4_3_2_Antiplatelet_Drugs

Anticoagulants

Pharmacological regulation of coagulation

The second main step in the stoppage of blood (haemo- stasis) is the clotting cascade. Have you ever baited a mouse with mouse and rat bait and wondered how they die? Well the bait is an anticoagulant that is used in warfarin. As you can understand it is very important to keep the dose of warfarin in the correct range, because too high a dose could result in haemorrhages which too little can result in excessive clotting. While there are newer rat baits and newer anticoagulants available, you can understand how important these medications are.

Video: The clotting cascade

Youtube video link.

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PHAR2001_4_3_7_ANTICOAGULANTS

Activity: Test your knowledge

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PHAR2001_4_3_3_Anticoagulants

Fibrinolytics

Overview of fibrinolysis

The final process involved in haemostasis, or the stoppage of blood, is actually the dissolving of the clot with a proteolytic (protein splitting) enzyme. That process is called fibrinolysis, and can be exploited with our medications, although, as you can imagine, there is a very fine line between the therapeutic dose and the one that can cause harm, or in other words, each of the fibrinolytic drugs have very narrow therapeutic index.

Video: fibrinolysis

Youtube video link.

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PHAR2001_4_3_7_FIBRINOLYTICS

Activity: Test your knowledge

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PHAR2001_4_3_4_Fibrinolytic_agents

Summary

Video: Summary of chapter

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4 - Thrombosis

Youtube video link.

viewed online inside Blackboard.

PHAR2001_4_3_8_CONCLUSION_THROMBOSIS

The Renal System Chapter 4: Diuretics

Introduction to diuretics

Objectives

The objectives for this chapter include:

- Describe the structure and function of the portions of the renal system relevant to diuretics.
- Identify and explain the actions interactions and side effects of diuretics.

The notes and PowerPoint for this chapter are directly below:

Video: Introduction to chapter 4: Diuretics

In this final chapter of Module 4, we explore the diuretics.

Youtube video link.

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PHAR2001_4_5_0_Intro_to_Diuretics

Understanding diuretics

Video: Understanding diuretics

In this lesson we review the structure and function of the

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kidneys.

Youtube video link.

PHAR2001_4_5_1_Understanding_Diuretics

Activity: Test your knowledge

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PHAR2001_4_5_1_Understanding_Diuretics

Understanding diuretics 2

The role of diuretics in heart failure and hypertension

The main indications for the use of diuretics are congestive heart failure and hypertension. In CHF, there is additional fluid being forced into the interstitial spaces due to increased back pressure. Relieving some of that fluid is beneficial as the heart can work more efficiently. In hypertension, recall that blood pressure is partly dependent on the amount of fluid, and as such, a reduction of fluid is likely to reduce BP - not dramatically, but reductions of 6-12 mmHg are common. Find out more of the foundational information about diuretics in this lesson.

Video: Understanding diuretics, part 2

Youtube video link.

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PHAR2001_4_5_2_Understanding_Diuretics_2

Activity: Test your knowledge

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PHAR2001_4_5_2_Understanding_Diuretics_2

Thiazide diuretics

Overview of Thiazide diuretics

The Thiazide and thiazide like diuretics are the most widely used diuretics. They are often combined with other diuretics to increase the efficacy and decrease the side effects. For instance, thiazide diuretics waste potassium. But if a small amount of thiazide diuretics is added along with a potassium sparing diuretic or to ACE inhibitors, or angiotensin receptor blockers, the potassium levels often normalise and the diuresis increases. Find more about the thiazide diuretics like hydrochlorothiazide and the thiazide like diuretics like metolazone in this lesson.

Video: Thiazide diuretics

Youtube video link.

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PHAR2001_4_5_3_Thiazide_Diuretics

Activity: Test your knowledge

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PHAR2001_4_5_3_Thiazide_Diuretics

Loop diuretics (not compulsory)

Video: Loop diuretics

The material in this lesson will not be assessed in PHAR2001.

Loop diuretics are very potent, reducing fluid volume more than most diuretics. It is the preferred diuretic for those people with Congestive Heart Failure (CHF).

Youtube video link.

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PHAR2001_4_5_4_Loop_Diuretics

Activity: Test your knowledge

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PHAR2001_4_5_4_Loop_Diuretics

Potassium sparing diuretics (Not compulsory)

Spironolactone and aldosterone receptor blockade

The material in this lesson will not be assessed in PHAR2001.

Aldosterone is a mineralocorticoid that is released from the adrenal cortex and is nicknamed the salt saver. It saves salt by binding to the cells in the distal convoluted tubule.

Spironolactone binds to the aldosterone receptors and blocks those receptors, preventing the salt saving actions. It also has antiandrogenic actions, blocking some of the testosterone

receptors.

- Although spironolactone is not as effective in reducing deaths in those individuals with CHF as compared to Loop diuretics, it is still highly rated for that.
- For hypertension, there are normally better diuretics, but in the subset of hypertensive individuals who have excess aldosterone (1% of the population), the medication is very effective.

Video: Potassium sparing diuretics

Youtube video link.

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PHAR2001_4_5_5_Potassium_Sparing_Diuretics

Activity: Test your knowledge

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PHAR2001_4_5_5_Potassium_Sparing_Diuretics

Other diuretics (not compulsory)

Alternative diuretics

The material in this lesson will not be assessed in PHAR2001.

The final lesson in the Cardiovascular and Renal system pharmacology has to do with the miscellaneous diuretics, the most important of which are the osmotic diuretics. These are not generally used to decrease fluid volume but rather they are used in special circumstances like intracranial hypertension and glaucoma. They are completely different to other types of diuretics and as such it is interesting to compare the actions and administration of these to other diuretics.

Video: Osmotic diuretics

Youtube video link.

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PHAR2001_4_5_6_Other_Diuretics

Activity: Test your knowledge

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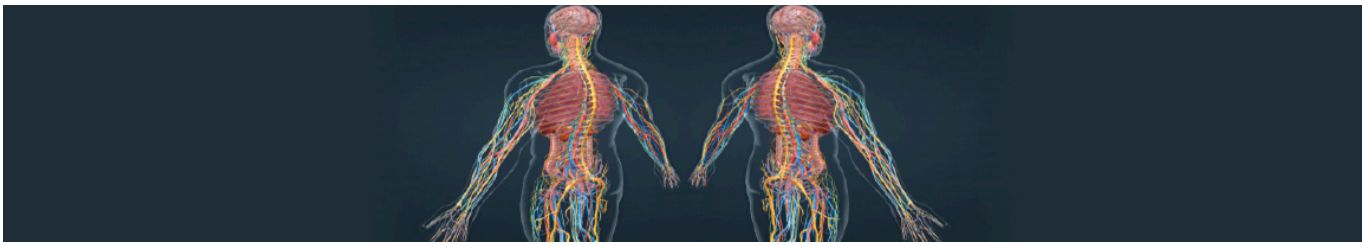
PHAR2001_4_5_6_Other_Diuretics

Module 5: Analgesics and Psychotropic Drugs



In this module, we explore analgesics and psychotropic drugs. The term 'psychotropic' pertains to one's mental state. The term 'analgesic' means pain killer. The way in which these drugs are related is that they both affect neurotransmitters (directly or indirectly).

Module overview



Overview of Module 5: Analgesics and psychotropic drugs

Welcome to Module 5 where we will be discussing the analgesics and the psychotropic drugs. As you probably already know, the analgesics are pain killers and the psychotropic drugs are drugs that affect behaviour, mood, thoughts or perception. While these two groups of medications initially seem like they would both be working at the level of the brain, most of the analgesics actually work somewhere along the pathways in which the pain is carried. Obviously, the stimuli that are carried along that pathway are ultimately perceived by the brain as pain.

These two topics are extremely important and include a disproportional amount of medications in the top 200 most prescribed. For instance, 12 of the top 100 most prescribed medications are antidepressants. When you are ready, click on the analgesics portion of this module and start furthering your understanding of pain.

By the end of this module you will be able to:

- Describe medications that act on the synthesis of the inflammatory mediators, including their indications, actions, side effects, and contraindications

- Discuss opioids with respect to the most significant contraindications and side effects.
- Define chronic pain and neuropathic pain and discuss the most common medications used to treat that type of pain
- Name and describe the most common pharmacological agents used in the treatment of Depression, Anxiety, Seizures, and Schizophrenia, including their indications, contraindications, mechanisms of action and side effects.

When you are ready, click on the analgesics portion of this module and start furthering your understanding of pain.

This entire module contains 2 hours of video lessons plus activities that reinforce that knowledge and prepare you for the final quiz.

Analgesics

Understanding Inflammation and pain introduction

The physiology and pathology of pain

Pain is a protective mechanism. It tells us when we need to tend to injuries, and helps further complications develop on those injuries. But when pain becomes chronic, there are permanent changes that magnify and perpetuate the pain. This has led to chronic pain now being considered a disorder in its own right. Given the importance of pain and the percentage of individuals living in chronic pain (20% of Australians), it is important that we learn as much about pain as we can.

Objectives

Objectives for Part 1

- Define the term ‘pain’ and briefly discuss the most important points about pain from a clinical aspect.
- Briefly describe inflammation, placing the most common anti-inflammatory agents into the context of that description.
- Describe the role of prostaglandins and cytokines in the process of inflammation and fever, and cite examples of their actions. Relate this to the anti-inflammatory agents that affect these mediators of inflammation.

- Briefly describe the pain pathway, overviewing the way in which our analgesics work.
- Describe the gate control theory of pain, and the way in which our understanding of pain has evolved.

You can find the PowerPoint and Notes for this Chapter directly below:

Video: Introduction to pain and analgesics

Youtube video link.

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PHAR2001_5_1_1_Introduction_to_Pain_and_Analgesics

Activity: Test your knowledge

Test yourself with these questions!

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PHAR2001_5_1_1_Introduction_to_Pain_and_Analgesics_ACTIVITIES

Mechanisms of inflammation and anti-inflammatory Agents

Understanding inflammation

Inflammation is an important contributor to the pain many clients experience because the process of inflammation sensitises the nerves and allows greater propagation of impulses that are ultimately perceived by the brain as pain. This lesson will help you picture the process of inflammation and the type of medications that are used to decrease inflammation.

Video: Mechanisms of inflammation

Youtube video link

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PHAR2001_5_1_2_Mechanisms_of_Inflammation_and_Anti-inflammatory_Agents

Activity: Test your knowledge

Reinforce your knowledge with these activities.

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Corticosteroids

Mechanisms of pain and analgesics

Predicting analgesic combinations by mechanism of action

Oftentimes, you will find that two medications combined will yield an additive effect where the client enjoys the benefits of both drugs. Normally that does not occur if the person takes two drugs that work by the same mechanism. With the information in this lesson, you will learn where our analgesics work so that you can more accurately predict which of the medications may yield additional benefits

Video: Mechanisms of pain and analgesics

Youtube video link.

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PHAR2001_5_1_3_Mechanisms_of_Pain_and_Analgesics

Activity: Test your knowledge

Questions to strengthen your active knowledge.

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PHAR2001_5_2_1_OTC_Intro_Fever_and_Antipyretics_ACTIVITIES

Intro to fever and antipyretics

Video: Intro to fever and antipyretics

https://www.youtube.com/watch?v=SKFzMtmTRS4

NSAIDs and COX2 inhibitors

In this lesson, we begin our understanding of the specific types of painkillers, with two groups of medications that actually work at the site of the pain. As tissues are damaged, the immune system cells come to the rescue, releasing cytokines, prostaglandins and other 'mediators of pain and inflammation'. Those are the target of the NSAIDs and COX2 inhibitors, which are discussed here.

Video: NSAIDs and COX2 inhibitors

Youtube video link

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PHAR2001_5_2_2_NSAIDs_and_COX2_Inhibitors

Activity: Test your knowledge

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PHAR2001_5_2_2_NSAIDs_and_COX2_Inhibitors_ACTIVITIES

Paracetamol (Acetaminophen)

Paracetamol: mechanisms, effects, and limitations

Contrary to the NSAIDs, Paracetamol, which is called Acetaminophen (Tylenol) in some countries does not have anti-inflammatory properties. (In this instance, we actually do need to know the other names of Paracetamol, because overseas clients will often be confused as to why their specific analgesic is not available in Australia.) Paracetamol does have anti-pyretic effects but only has only a fraction of the amount of analgesic effects as compared to NSAIDs. This brings up an important point, because most people think that, because it is an analgesic, if they are not getting the pain relief that they expect from the first dose, they should just take more. But a weak analgesic does not get more effective with a greater dose – there is just more toxic effects on the liver! Paracetamol toxicity is the number one cause of acute liver damage in Australia, so we should know as much about paracetamol as possible.

Video: Paracetamol Acetaminophen

Youtube video link.

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PHAR2001_5_2_3_Paracetamol_Acetaminophen

Activity: Test your knowledge

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PHAR2001_5_2_3_Paracetamol_Acetaminophen

Pain relief in children and elderly

Pain management for the elderly and children

In this lesson, we will be learning about the management of pain in children and the elderly. Both of those groups of people have a decreased ability to metabolise and excrete medications. But in addition, there are other specific issue for both groups. For instance, one would normally think that if the dose for a 100 kg adult is 100 mg, the 10 kg child should take 10 mg. But that dose could cause serious harm or death of the child, because dose for children is not calculated that way. For the elderly, they are often on several medications at the same time so that, combined with their inability to metabolise the medications quickly will lead to special issues. Let's explore the specific issues for these clients in more depth.

Video: Pain relief in children and the elderly

Youtube video link.

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PHAR2001_5_2_4_Pain_Relief_in_Children_and_Elderly

Activity: Test your knowledge

Questions to reinforce your knowledge!

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PHAR2001_5_2_4_Pain_Relief_in_Children_and_Elderly

Opioid overview

The use of opioids

Opium is the exudate that comes from the opium poppy plant and contains morphine and codeine. Up until the early 1960s there was a medication called paregoric that contained opium and was sold to reduce diarrhoea, reduce cough and decrease pain. But it was also used in many households to settle down the children before going to sleep! Times have changed, and we no longer use opium or use opioids when our young children are over-excited (or pretty much for any reason), but the opioids remain a very important group of medications for adults. Let's learn more about their present use.

Video: Opioid overview

Youtube video link.

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PHAR2001_5_3_1_Opioid_Overview

Activity: Test your knowledge

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PHAR2001_5_3_1_Opioid_Overview

Opioid mechanisms

Video: Opioid mechanisms

The list of side effects that can be experienced goes on and on, but it

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is not enough just to be able to rattle off that list. By understanding how opioids work and why they cause the side effects, we can better instruct clients who are being administered those medications.

Youtube video link.

PHAR2001_5_3_2_Opioid_Mechanisms

Activity: Test your knowledge

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PHAR2001_5_3_2_Opioid_Mechanisms

Specific opioids (Not compulsory)

Additional types of opioids

The material in this lesson will not be assessed in PHAR2001.

We have 17 different types of opioid receptors in the body, and their prevalence at various points in the body vary with each receptor. Meanwhile, our opioid medications affect each of those receptors with varying amounts of affinity. And to make it even more complicated, some of the opioids are agonists at one receptor and antagonists at others. It would take an entire unit of study to get a really good picture of all of the opioids, but in this unit, we first need to understand that we use 10mg IV morphine as the reference point and compare all drugs to that. And then we will take just a few of the most highly prescribed opioids and compare them to morphine.

Video: Methadone (Not

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compulsory)

Youtube video link.

PHAR2001_5_3_3_Specific_Opioids

Activity: Test your knowledge

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PHAR2001_5_3_3_Specific_Opioids

Chronic and neuropathic pain

Treatment of chronic and neuropathic pain

It is interesting to note that the analgesics like opioids that are very effective at reducing extreme, acute pain are generally not effective once the pain becomes chronic. That type of pain has resulted in changes along the pain pathway that magnify the pain and make the pain worse, so it needs to be treated with different medications. This lesson will give you some more insight into the treatment of chronic and neuropathic pain.

Video: Chronic and neuropathic pain

Youtube video link.

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PHAR2001_5_4_1_Chronic_and_Neuropathic_Pain

Activity: Test your knowledge

Engage in these activities to fully understand the material.

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PHAR2001_5_4_1_Chronic_and_Neuropathic_Pain

Thank you for your participation in this portion of the module!

Thanks for your participation in this chapter. Pain and treatment of pain are so important that chronic pain is considered a disorder in its own right. It is important that we make sure that we do everything possible to treat pain before it becomes chronic, and to learn as much about the treatment of pain as possible.

Psychotropic Medications

Introduction to psychotropic medications

Introduction to Psychotropic drugs

Psychotropic drugs are drugs that affect behaviour, mood, thoughts or perception.

We will start out the psychotropic drugs by getting a good picture of the neurodegenerative disorders. While there are several types of progressive neurodegenerative disorders, over 90% of cases are either Parkinson's disease or Alzheimer's disease, so the treatment of those disorders will be our focus.

The objectives for this chapter include:

- For each of the disorders listed below, briefly explain the clinical manifestations of the disorder, the hypothesised neurologic abnormalities, and the mechanisms by which the commonly prescribed medications help correct the neurologic abnormality.
  - Parkinson’s disease
  - Alzheimer’s disease
  - Anxiety disorders,
  - Depression,
  - Schizophrenia and other psychotic disorders and

    - Seizure syndromes

- And for each of the medications discussed, note any associated suffix, and explain the side effects and contraindications

You can access the Notes and PowerPoint for this chapter here:

Medications for Parkinson's disease

Treating Parkinson's disease

Parkinson’s disease is a chronic, progressive degenerative disorder that at this stage has no universal cure. Unfortunately, our medications can give a higher quality of life to clients but will not reverse or stop the progression of the disease. There are several different treatments of Parkinson’s and this lesson concentrates on the two most established treatments.

Video: Medication for Parkinson's disease

Youtube video link.

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PHAR2001_5_5_1_Parkinsons_Medications

Activity: Test your knowledge

Learn more about the treatment of Parkinson's disease with these activities.

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PHAR2001_5_5_1_Parkinsons_Medications

Medications for treatment of Alzheimer's disease

Current treatment of Alzheimer's disease

Alzheimer’s disease is another chronic, progressive degenerative disorder, but it involves completely different neurons in different portions of the brain. Unlike Parkinson’s disease, there are few changes in movement until late in the course of Alzheimer’s disease. But similar to Parkinson’s disease, we have not found medications to alter the course of Alzheimer’s disease for more than a few months. Let’s take a look at this very important disorder and its treatment.

Video: Watch this video about Alzheimer disease medications

Youtube video link.

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PHAR2001_5_5_2_Alzheimers_Medications

Activity: Test your knowledge

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PHAR2001_5_5_2_Alzheimers_Medications

Anxiety

Benzodiazepines

Right now, we are going to explore the Anxiolytics and hypnotics. Anxiolytics are medications that reduce anxiety and hypnotics are medications that help the client sleep. Often the anxiolytics are useful hypnotics. The benzodiazepines are probably the most important

groups of medications. You have probably heard of Zanax (Alprazolam) and/or Valium (Diazepam). Those are two benzodiazepines that are in the top 200 most prescribed medications but Clonazepam and Lorazepam are also benzodiazepines that are highly prescribed. The benzo's as you may nickname them in the future are variable with respect to the half-life and often the exact types of disorders that they have been proven to treat. For instance, some like alprazolam have antidepressant actions, and a half-life of 11 hours. While that may seem long, others such as Diazepam have a much longer half-life of 50 hours. Remember, the longer the half-life the greater risk of developing toxicity, so it is important to look out for that with the longer-acting Benzos.

Video: Benzodiazepines

Youtube video link.

NOTE: Just before the end of the video, I say that the benzodiazepines are a schedule 8 drug. I meant to say the benzodiazepines flunitrazepam and alprazolam are a schedule 8 and the others are a schedule 4. Sorry for not being clear. Regards, Ken

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PHAR2001_5_6_1_Benzodiazepines

Activity: Test your knowledge

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PHAR2001_5_6_1_Benzodiazepines

Alternatives to benzodiazepines

The need for alternative medications

Because of the fact that the benzodiazepines are addictive, are often abused, and are associated with an increased risk of developing Alzheimer's disease, we are always eager to see alternative treatments for anxiety. Buspirone is not a benzodiazepine. But it is ineffective for panic anxiety, as it is mostly approved for just generalised anxiety. Another interesting medication that is mainly used for children is one of the first-generation antihistamines called Hydroxyzine. Recall that the first-generation antihistamines (like diphenhydramine), broke into the BBB and resulted in drowsiness.

Video: Alternatives to benzodiazepines

Youtube video link.

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PHAR2001_5_6_2_Alternatives_to_Benzodiazepines

Activity: Test your knowledge

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PHAR2001_5_6_2_Alternatives_to_Benzodiazepines

Depression

Introduction to Depression

We are going to shift our attention to depression right now. Depression is a very common disorder that is very much linked to stress, as stress seems to dysregulate the area of the brain associated with mood. The present hypothesis is that the person who is depressed normally has a (relative) decrease in the levels level of serotonin and/or noradrenaline in the

areas of the brain associated with mood. Therefore, our antidepressants mostly concentrate on increasing one or both of those neurotransmitters. There are a few ways in which we can do that, as we will see in this next video.

Video: Introduction to depression

Youtube video link.

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PHAR2001_5_7.1_Introduction_to_Depression

Activity: Test your knowledge

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PHAR2001_5_7.1_Introduction_to_Depression

SSRIs and TCAs

The use of SSRIs in managing depression

The SSRIs (selective serotonin reuptake inhibitors are the first line agents in the treatment of major depression. There are more SSRIs in the top 200 most prescribed medications than we will need to learn but you really need to remember the ones in the top 20 most prescribed, which are sertraline (Zoloft), Escitalopram (Lexapro) and Fluoxetine (Prozac). These medications block the re-uptake of the serotonin, leaving the serotonin in the synaptic cleft longer to perform its inhibitory action on the next neuron. Lets take a look at this very important group of medications.

Video: SSRIs and TCAs

Youtube video link.

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PHAR2001_5_7_2_SSRIs_and_TCAs

Activity: Test your knowledge

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PHAR2001_5_7_2_SSRIs_and_TCAs

SNRIs and atypical antidepressants

Introduction to SNRIs

While SSRIs are the most commonly prescribed group of antidepressants, there is a newer group of medications that works in a similar way to the SSRIs but affects both serotonin and noradrenaline. The medication is called an SNRI (Serotonin Noradrenaline Reuptake Inhibitor). The side effect profile is similar, but in addition to the antidepressant effects, the SNRIs may have an effect reducing chronic and neuropathic pain. There are also what we call 'atypical antidepressants' that tend to work at multiple sites.

Video: Antidepressants

Youtube video link.

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PHAR2001_5_7_3_SNRIs_and_Atypical_Antidepressants

Activity: Test your knowledge

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PHAR2001_5_7_3_SNRIs_and_Atypical_Antidepressants

Psychosis (Not compulsory for this unit)

Introduction to antipsychotics

The material in this lesson is not compulsory in PHAR2001.

Treatment of psychosis is confusing but it is easiest to start out by understanding the dopamine pathways because treatment involves decreasing the dopamine in the brain. In the diagram below, the blue arrows depict the dopamine pathways. The mesolimbic dopamine pathway is the one that is therapeutic. People with schizophrenia have a hyperactive limbic system, and when we give dopamine antagonists, the positive symptoms (like delusions, disorganised speech and/or thought, distorted perceptions and emotions, hallucinations and the belief that one is under the control of an external agency). That is good, and that is what we want from a medication that treats psychosis.

Video: Antipsychotics (Not compulsory for this unit)

Youtube video link.

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PHAR2001_5_8_1_Intro_to_Antipsychotics

Activity: Test your knowledge

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PHAR2001_5_8_1_Intro_to_Antipsychotics

Specific antipsychotics (Not compulsory for this unit)

Introduction to specific antipsychotics

The material in this lesson is not compulsory in PHAR2001.

The antipsychotic medications can be divided into the typical (first generation) and the atypical (second generation) antipsychotics. While both of those groups of medications share the same therapeutic goal, namely to block the dopamine 2 in the brain, the typical antipsychotics block all dopamine pathways about equally while the atypical (second generation) antipsychotics are more selective for one pathway or another.

Video: Specific antipsychotics (Not compulsory for this unit)

Youtube video link.

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PHAR2001_5_8_2_Specific_Antipsychotics

Activity: Test your knowledge

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PHAR2001_5_8_2_Specific_Antipsychotics

Seizures (Not compulsory for this unit)

Antiseizure medications

The material in this lesson is not compulsory in PHAR2001.

This last lesson deals with anti-seizure medications. A seizure is defined as a sudden and uncontrollable surge of electrical activity in the brain. There are several different types of seizures, but the most important thing is to distinguish between a partial (or focal) seizure which does not cross into both hemispheres of the brain and a generalised seizure, which does cross into the other side of the brain, with the client losing consciousness. When a person gets recurrent seizures of either type, the person is said to have epilepsy. Antiseizure medications need to break into the blood brain barrier and decrease positive ions (Sodium or calcium) in the brain or increase negative ions (chloride) in the brain. As you would expect, this would normally result in sleepiness and possibly a bit of mental confusion, but nonetheless, seizures are very important to treat.

Video: Anti-seizure medications (Not compulsory for this unit)

Youtube video link.

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PHAR2001_5_8_3_Antiseizure_Medications

Activity: Test your knowledge

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PHAR2001_5_8_3_Antiseizure_Medications

Well done: Congratulations!

As you have seen, the psychotropic medications were challenging to understand, but you got through it and you are on the home stretch. Once you complete the activities in the workbook for Module 5 you can proceed to our final module, which deals with the medications that deliver selective toxicity. See you there!

Module 6: Medications with Selective Toxicity



Remember how our medications normally use our own cellular receptors or our own enzymes? Well our medications that combat infections or cancer will need to take a different approach. Let's find out how they do that.

Module overview

Introduction to Module 6

This module has only 43 minutes of compulsory lessons. You may choose to engage in just the compulsory activities (choose the first chapter below) or choose the entire chapter which includes examples of specific antibiotics, antifungal agents, antiviral agents and chemotherapeutic agents by choosing the second chapter below. The third chapter is an interesting artistic reflection of the unit which is not compulsory.

Video: Module overview - Medications with selective toxicity

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PHAR2001_6_0_1_Introduction_to_Module_6

Shortened Chapter: Only the 43 minutes of compulsory lessons on Medications with Selective Toxicity

A/1: Antibiotics

Chapter 1: Medications with selective toxicity

The Notes and PowerPoint for this chapter, inclusive of the non-compulsory portions are directly below.

Selective toxicity module 6 PPT.pptx

Introduction to Antibiotics: Definitions

In this first topic in Module 6, we are going to get an understanding of the terms used when discussing antimicrobial agents and anti-cancer chemotherapy and also make sure that you have a good picture of the difference between viruses, bacteria and fungi.

Video: Antibiotics

Youtube video link.

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PHAR2001_6_1_1_Antibiotics_Definitions

Activity: Test your knowledge

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PHAR2001_6_1_1_Antibiotics_Definitions

Antibiotics overview

Video: Antibiotics overview

We will be continuing our general discussions of antibiotics in this lesson, getting an understanding of why certain side effects are common; how to avoid contributing to antibiotic resistance, and understanding the way in which a specific antibiotic is selected.

Youtube video link.

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PHAR2001_6_1_2_Antibiotics_Overview

Activity: Test your knowledge

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PHAR2001_6_1_2_Antibiotics_Overview

C/3: Antifungal agents

A brief look at antifungal agents

Fungi are nature's way of degrading things, so if you want your belongings to last longer, make sure that you clean off the mould and mildew!

Fungi are found all over, and we have an astounding number of fungi that live in the natural skin flora, but the bacterial toxins from the resident bacteria keep the fungi in check. When an infection does occur from fungi, the term is a mycosis. That is where our antifungal infections come in, and unfortunately, because of the fact that fungal cells are so close to human cells, there is potential toxicity with most antifungal agents. The main difference between human cells and the fungal cells is that human cells contain cholesterol in the cell membrane, while fungal cell membranes contain ergosterol. So the ergosterol is the target of most of our antifungal agents.

Video: Antifungal agents

Youtube video link.

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PHAR2001_6_3_1_Antifungal agents shortened

Activity: Test your knowledge

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PHAR2001_6_3_1_Antifungal agents

D/4: Antiviral agents

Antiviral agents

We knew very little about viruses until the HIV scare in the early 1980s but since then our understanding of viruses has been advancing comparatively quickly. As such, the first-line therapies for viral infections will change before you get to clinic. So the good news is that you do not need to know many antiviral medications at all. But you really do need to have a general knowledge of viruses and of the antiviral medications. The next two lessons attempt to give you that broad understanding.

Video: Antiviral medications

Youtube video link.

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PHAR2001_6_4_1_Viruses 1 shortened

Activity: Test your knowledge

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PHAR2001_6_4_1_Viruses_1

E/5: Anticancer chemotherapy

Anticancer Chemotherapy

Normal cells are tightly regulated, and this ensures that they do not start proliferating out of control. When a group of cells lose these controls, they may become cancerous. In this lesson, we are going to take a general look at the types of anticancer chemotherapeutic agents and develop an understanding of why some of the common side effects from these agents develop.

Video: Anticancer medications

Youtube video link.

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PHAR2001_6_5_1_Anticancer_Chemotherapy_shortened

Activity: Test your knowledge

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PHAR2001_6_5_1_Anticancer_Chemotherapy

Thank you for your participation!

Thanks for engaging in the activities associated with the medications that deliver selective toxicity. Our final topic is just a reflective piece that we really hope you enjoy.

Medications with Selective Toxicity (Full chapter, including non-compulsory lessons and activities)

A/1: Antibiotics

Introduction to chapter 1

The Notes and PowerPoint directly below contain the objectives for all parts of this chapter.

Selective toxicity module 6 PPT(1).pptx

Introduction to Antibiotics: Definitions

In this first topic in Module 6, we are going to get an understanding of the terms used when discussing antimicrobial agents and anti-cancer chemotherapy and also make sure that you have a good picture of the difference between viruses, bacteria and fungi.

Video: Antibiotics

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PHAR2001_6_1_1_Antibiotics_Definitions

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_6_1_1_Antibiotics_Definitions

Antibiotics overview

Video: Antibiotics overview

We will be continuing our general discussions of antibiotics in this lesson, getting an understanding of why certain side effects are common; how to avoid contributing to antibiotic resistance, and understanding the way in which a specific antibiotic is selected.

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PHAR2001_6_1_2_Antibiotics_Overview

Activity: Test your knowledge

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PHAR2001_6_1_2_Antibiotics_Overview

B/2: Specific antibiotics

Specific Antibiotics 1

Our antibiotics can be grouped in many different ways. For instance, one grouping has them as time dependant versus dose dependent. But one of the most useful ways to group them is by mechanism of action. While we will not need to know the mechanism by which various types of antibiotics work, it is a convenient grouping that ultimately makes it easier to get a complete picture of each type of antibiotic. In this first lesson about specific antibiotics, we will look at the antibiotics that inhibit cell wall synthesis. This happens as the bacteria are reproducing, and the prodigy will be unable to make the cell wall needed to survive. Antibiotics that inhibit the cell wall synthesis include the penicillins (suffix: cillin) and the cephalosporins (prefix: cef or ceph). These contain a β (beta)-lactam ring that is the active portion of the drug. Bacteria often develop resistance to β-lactam antibiotics by synthesizing a β-lactamase, an enzyme that attacks the β-lactam ring before it can adversely affect the cell wall. To overcome this resistance, β-lactam antibiotics can be given with β-lactamase inhibitors such as clavulanic acid.

Video: Specific antibiotics 1

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PHAR2001_6_2_1_Specific_Antibiotics_1

Activity: Test your knowledge

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PHAR2001_6_2_1_Specific_Antibiotics_1

Specific Antibiotics 2

Specific antibiotics continued

In this lesson, we broaden our understanding of ways in which medications can kill bacteria with antibiotics that inhibit DNA, block protein synthesis and inhibit metabolic processes.

For instance, Sulphonamides including sulphamethoxazole inhibits the ability of the bacteria to synthesise its folic acid that is needed to reproduce. Those medications are bacteriostatic, keeping the microorganisms in check while the immune system gains control.

Video: Specific antibiotics 2

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PHAR2001_6_2_2_Specific_Antibiotics_2

Activity: Test your knowledge

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PHAR2001_6_2_2_Specific_Antibiotics_2

C/3: Antifungal agents

A brief look at antifungal agents

Fungi are nature's way of degrading things, so if you want your belongings to last longer, make sure that you clean off the mould and mildew!

Fungi are found all over, and we have an astounding number of fungi that live in the natural skin flora, but the bacterial toxins from the resident bacteria keep the fungi in check. When an infection does occur from fungi, the term is a mycosis. That is where our antifungal infections come in, and unfortunately, because of the fact that fungal cells are so close to human cells, there is potential toxicity with most antifungal agents. The main difference between human cells and the fungal cells is that human cells contain cholesterol in the cell membrane, while fungal cell membranes contain ergosterol. So the ergosterol is the target of most of our antifungal agents.

Video: Fungi and antifungal agents

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PHAR2001_6_3_1_Antifungal agents

Activity: Test your knowledge

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PHAR2001_6_3_1_Antifungal agents

D/4: Viruses 1

Viruses 1

We knew very little about viruses until the HIV scare in the early 1980s but since then our understanding of viruses has been advancing comparatively quickly. As such, the first-line therapies for viral infections will change before you get to clinic. So the good news is that you do not need to know many antiviral medications at all. But you really do need to have a general knowledge of viruses and of the antiviral medications. The next two lessons attempt to give you that broad understanding.

Video: Antiviral medications

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PHAR2001_6_4_1_Viruses_1

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_6_4_1_Viruses_1

Viruses 2

Video: Antiviral medications

In this lesson, we will continue to develop our understanding of the antiviral medications, learning about the medications that are used to treat infections caused by the

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PHAR2001_6_4_2_Viruses_2

herpes strain of viruses, the HIV and the hepatitis-causing viruses.

Activity: Test your knowledge

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PHAR2001_6_4_2_Viruses_2

E/5: Anticancer chemotherapy

Anticancer Chemotherapy

Normal cells are tightly regulated, and this ensures that they do not start proliferating out of control. When a group of cells lose these controls, they may become cancerous. In this lesson, we are going to take a general look at the types of anticancer chemotherapeutic agents and develop an understanding of why some of the common side effects from these agents develop.

Video: Anticancer medications

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PHAR2001_6_5_1_Anticancer_Chemotherapy

Activity: Test your knowledge

This item is an interactive component that cannot be exported. This item must be viewed online inside Blackboard.

PHAR2001_6_5_1_Anticancer_Chemotherapy

Thanks for your participation!

Thanks for engaging in the activities associated with the medications that deliver selective toxicity. Our final topic is just a reflective piece that we really hope you enjoy.

Polypharmacy 1: A reflective section

Introduction

Hi All,

This is the very last chapter of the unit, and it is just a reflective piece for you to think about the potential over-reliance on medications.

Hope you enjoy! No notes or PPT for these two videos.

Best wishes,

Ken

Videos: Overview of the reliance on medications

https://www.youtube.com/watch?v=JNihp-m2Um0

https://www.youtube.com/watch?v=Lp3pFjKoZI8

Goodbye for now.

Hi All,

Well thank you for your participation in the unit. I really hope that you enjoyed it and learned some valuable skills in the process.

Do make a great future for yourself!

Warm regards and best wishes,

Ken

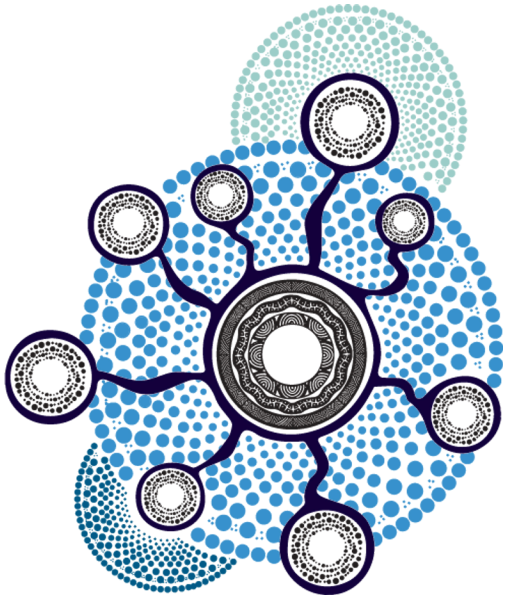
Acknowledgement of Traditional Custodians

We acknowledge and pay respect to the ancestors and descendants of the Lands upon which we meet and study. We are mindful that within and without the concrete and steel of buildings, the Land always was and always will be Aboriginal Land. We also acknowledge and pay respect to all students studying this Unit as we walk together through this session's learning journey.

Acknowledgement of Origins Artwork

We acknowledge Widjabal Country as the Country where Southern Cross University originated and grew, and from where it continues to grow. It seeded the growth of our Gumbaynggirr Campus in Coffs Harbour, our Yugambeh Campus in Coolangatta and it planted the seeds to transform the Countries we are on.

Understanding our origins will make each step into the future a more purposeful one, leading all to transform tomorrow.



Artist: Sheldon (SJ) Harrington

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